

CSE 4621

Machine Learning

Topic: Neural Networks

Source: [DeepLearning.AI](https://deeplearning.ai)

Presented by

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Assistant Professor



Neural networks

Origins: Algorithms that try to mimic the brain.

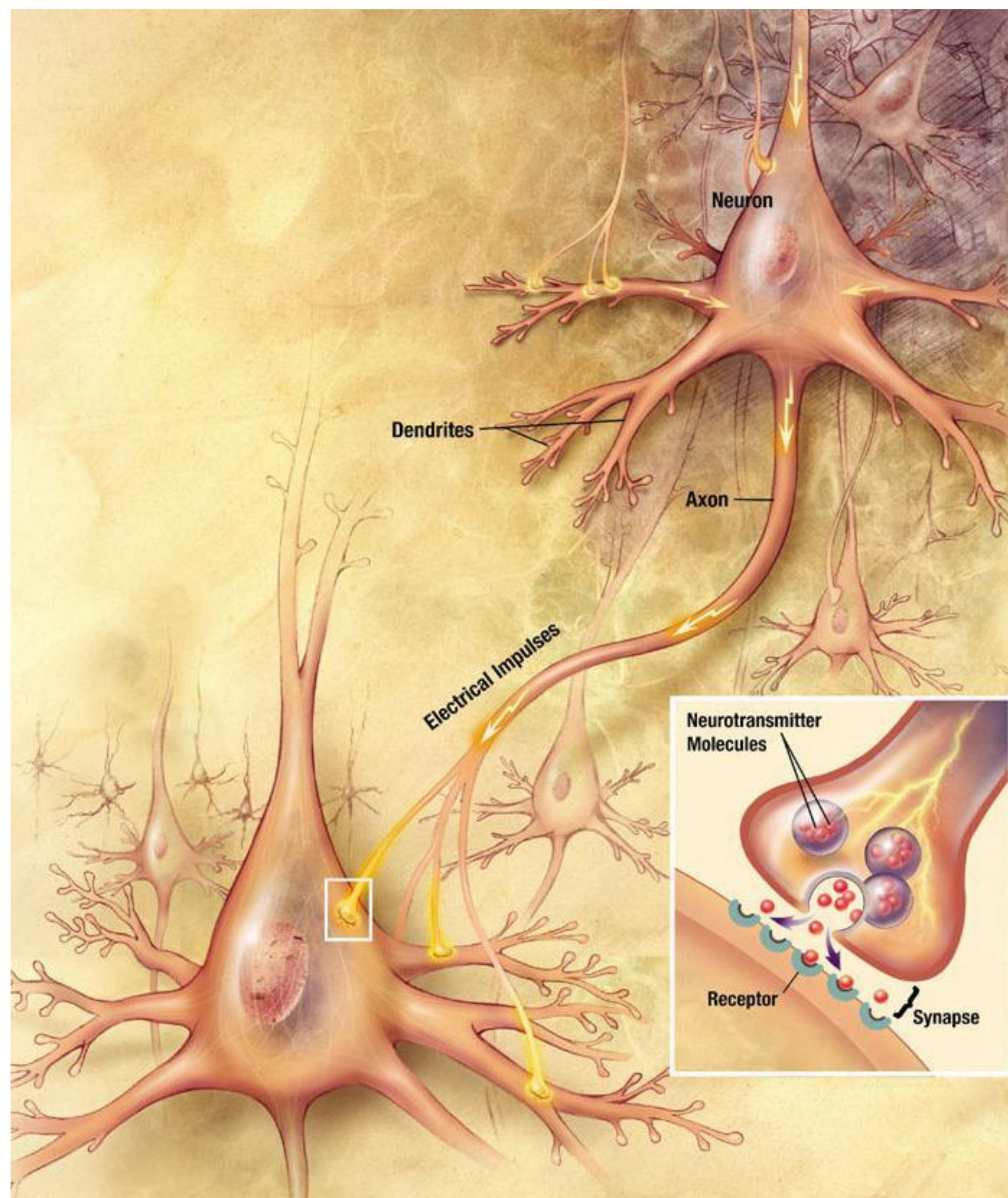
Used in the 1980's and early 1990's.
Fell out of favor in the late 1990's.

Resurgence from around 2005.

speech → images → text (NLP) → ...



Neurons in the brain

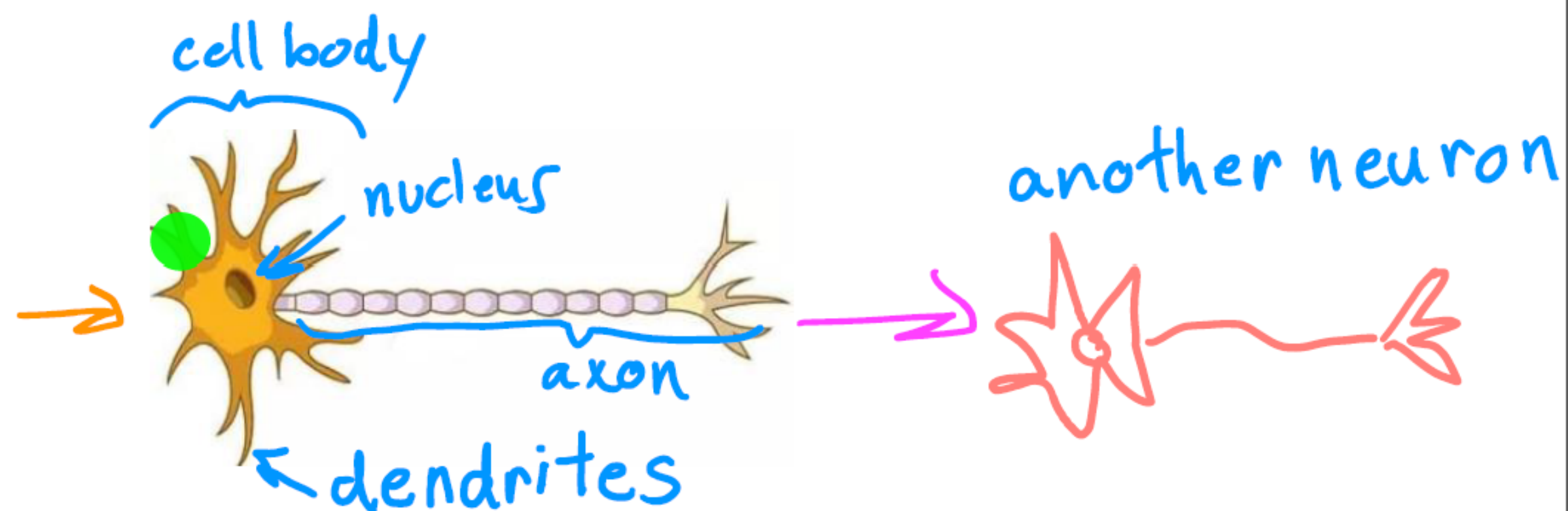


Neural networks

Biological neuron

inputs

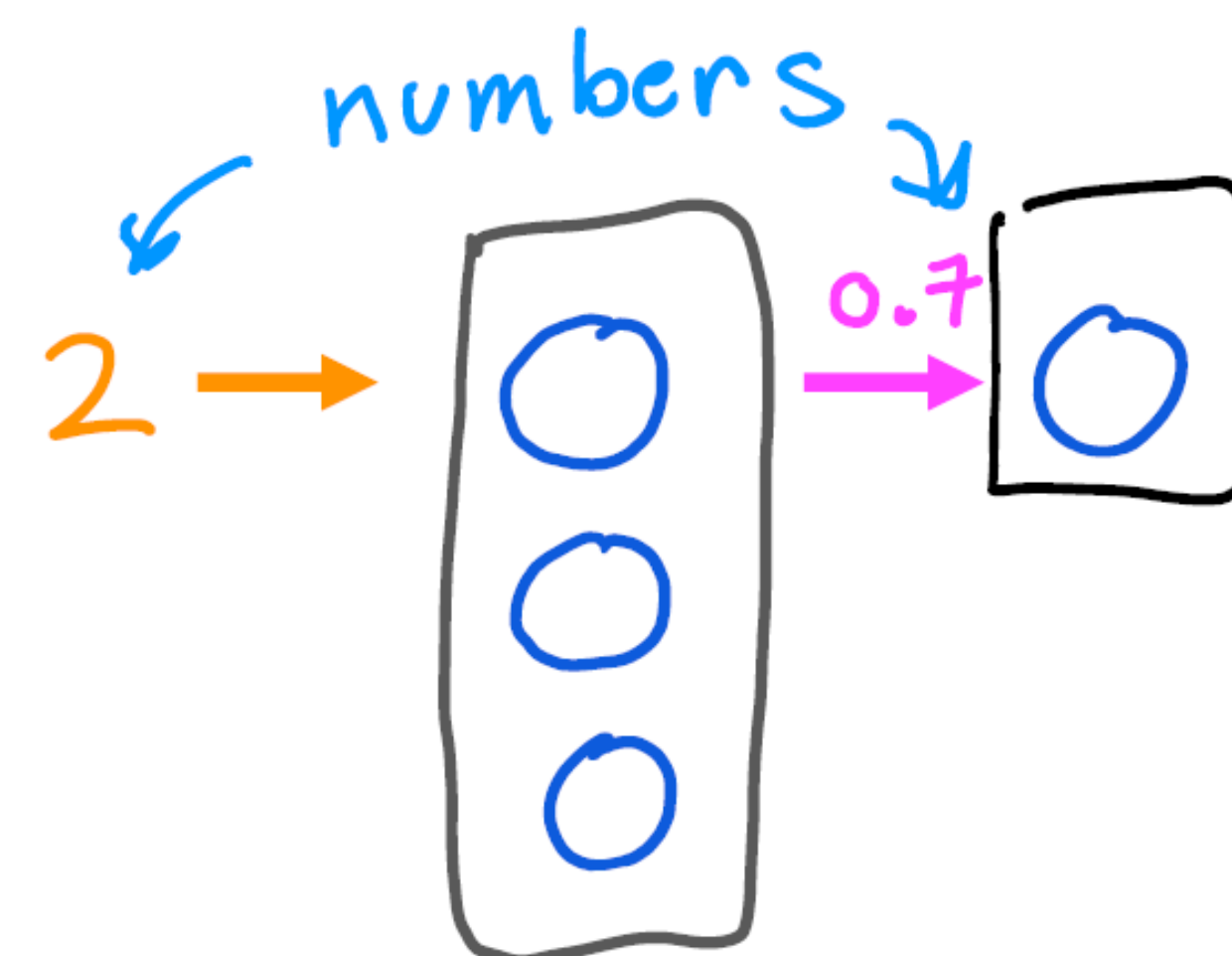
outputs



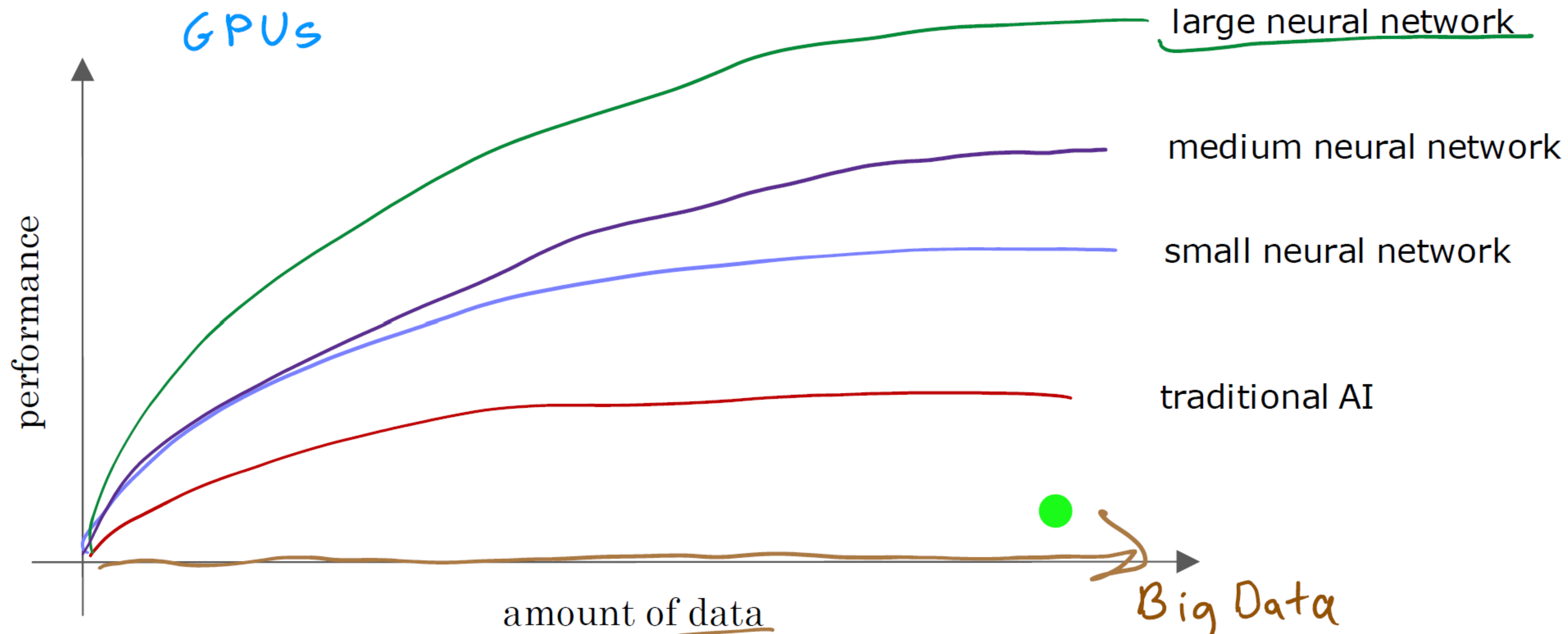
Simplified mathematical model of a neuron

inputs

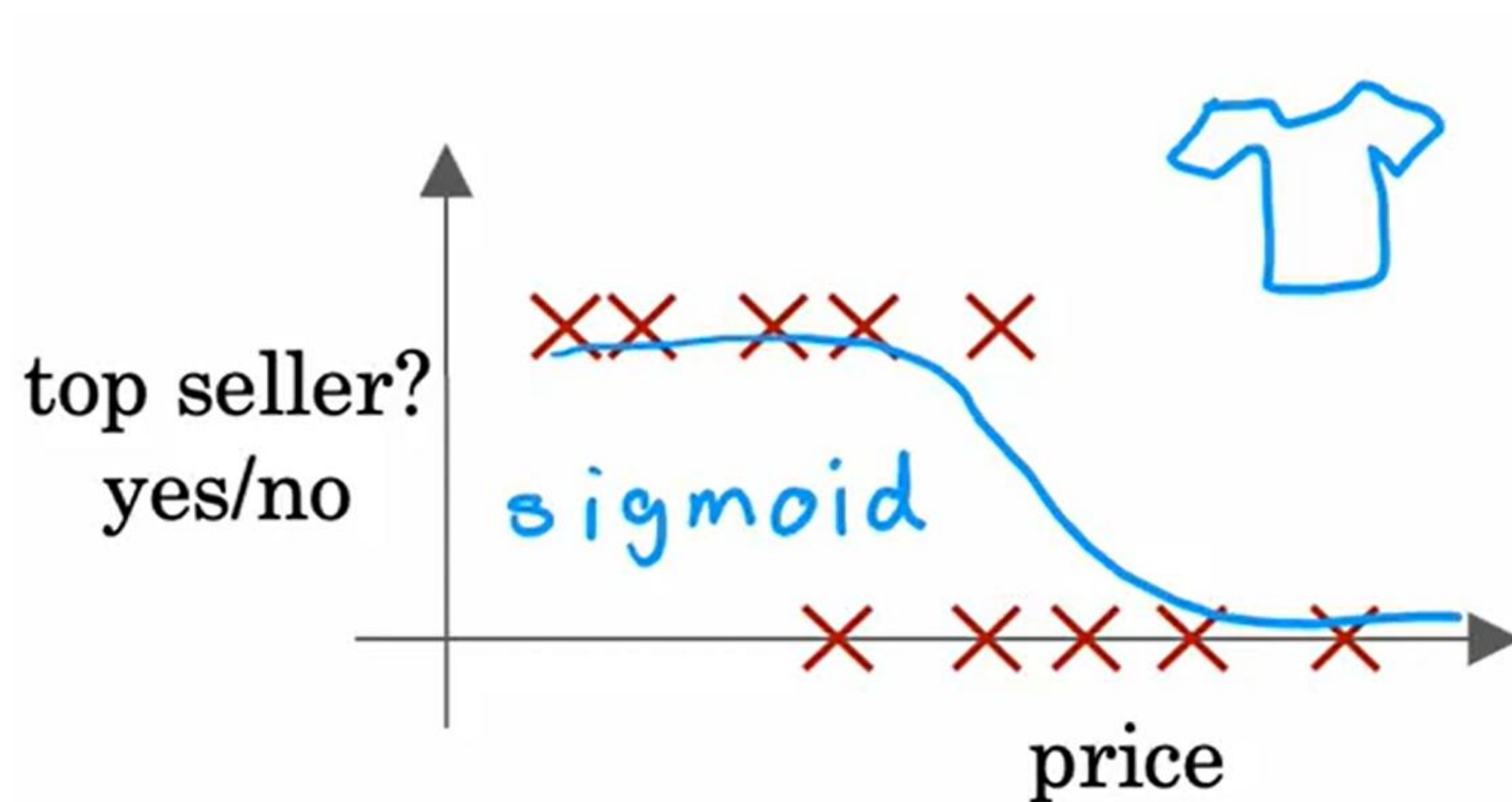
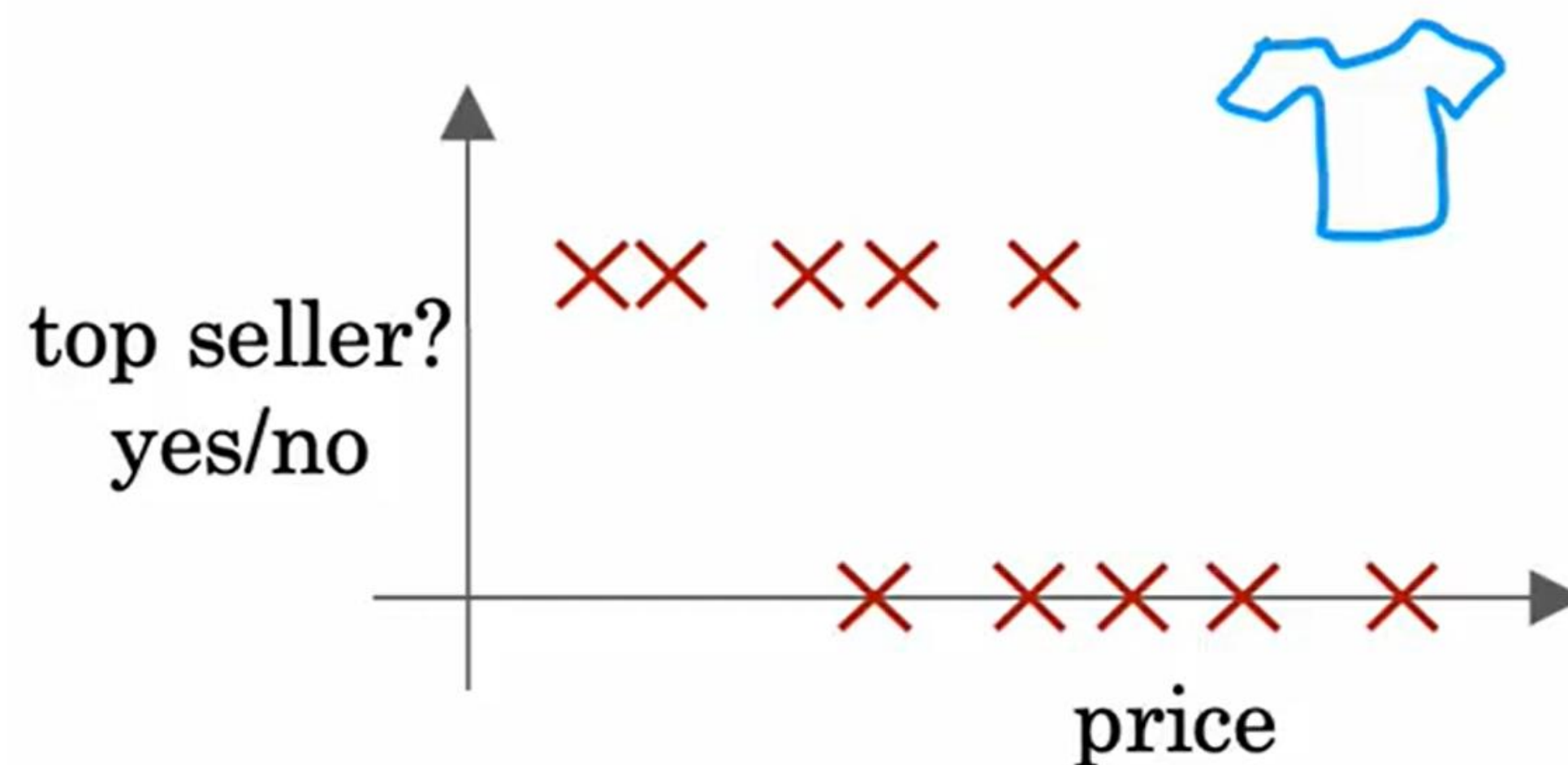
outputs



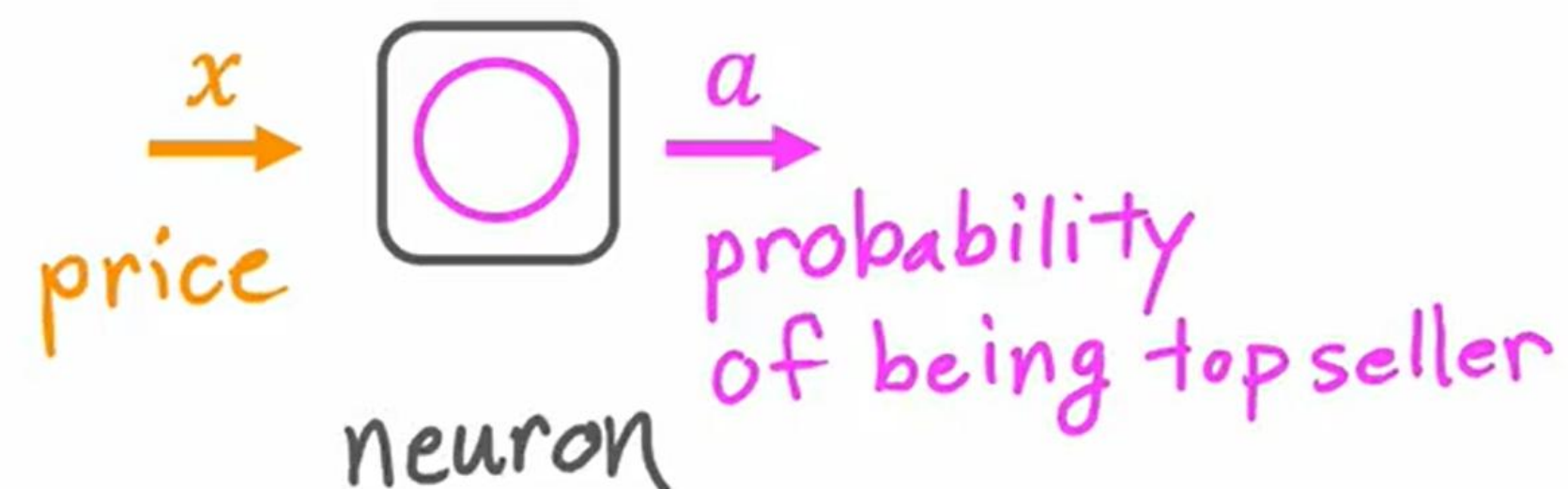
Why Now?



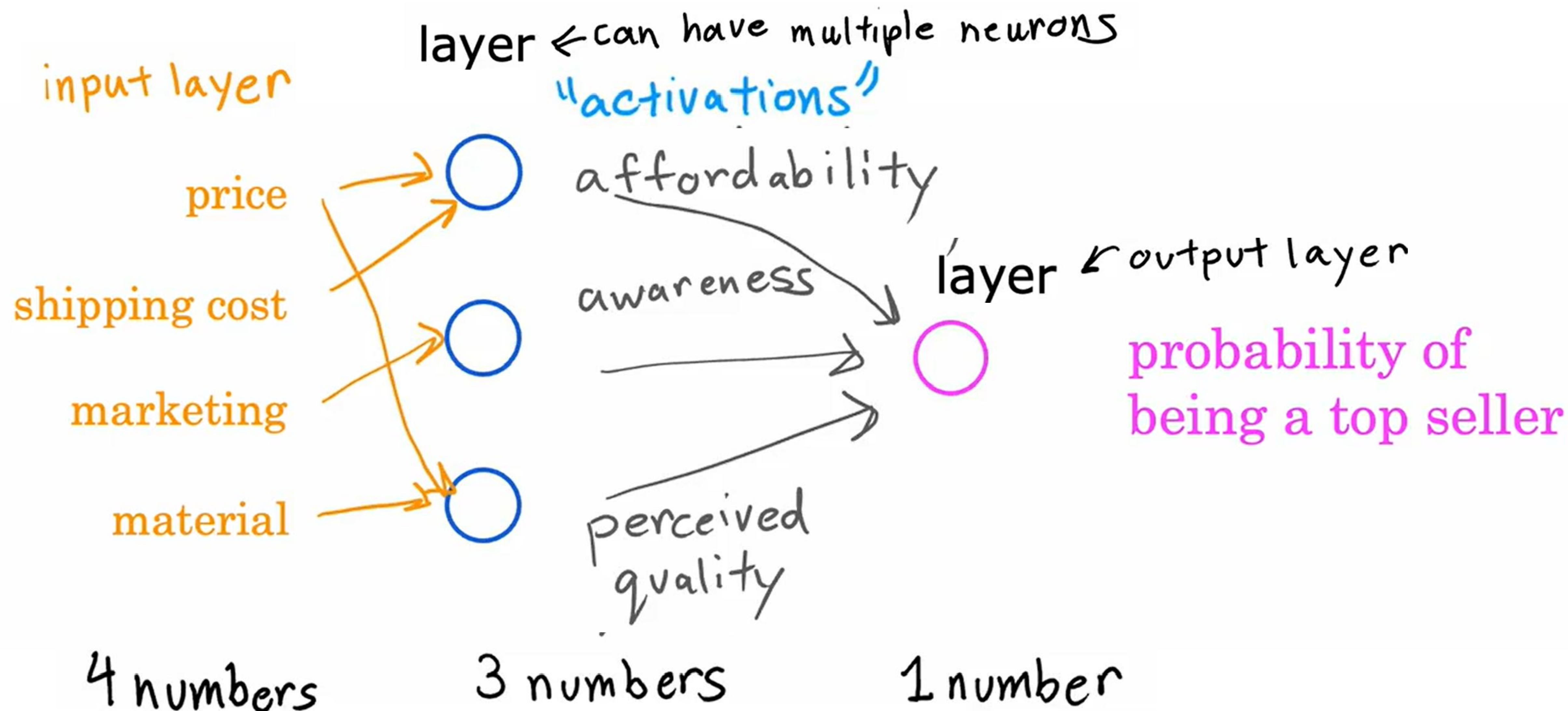
Demand Prediction



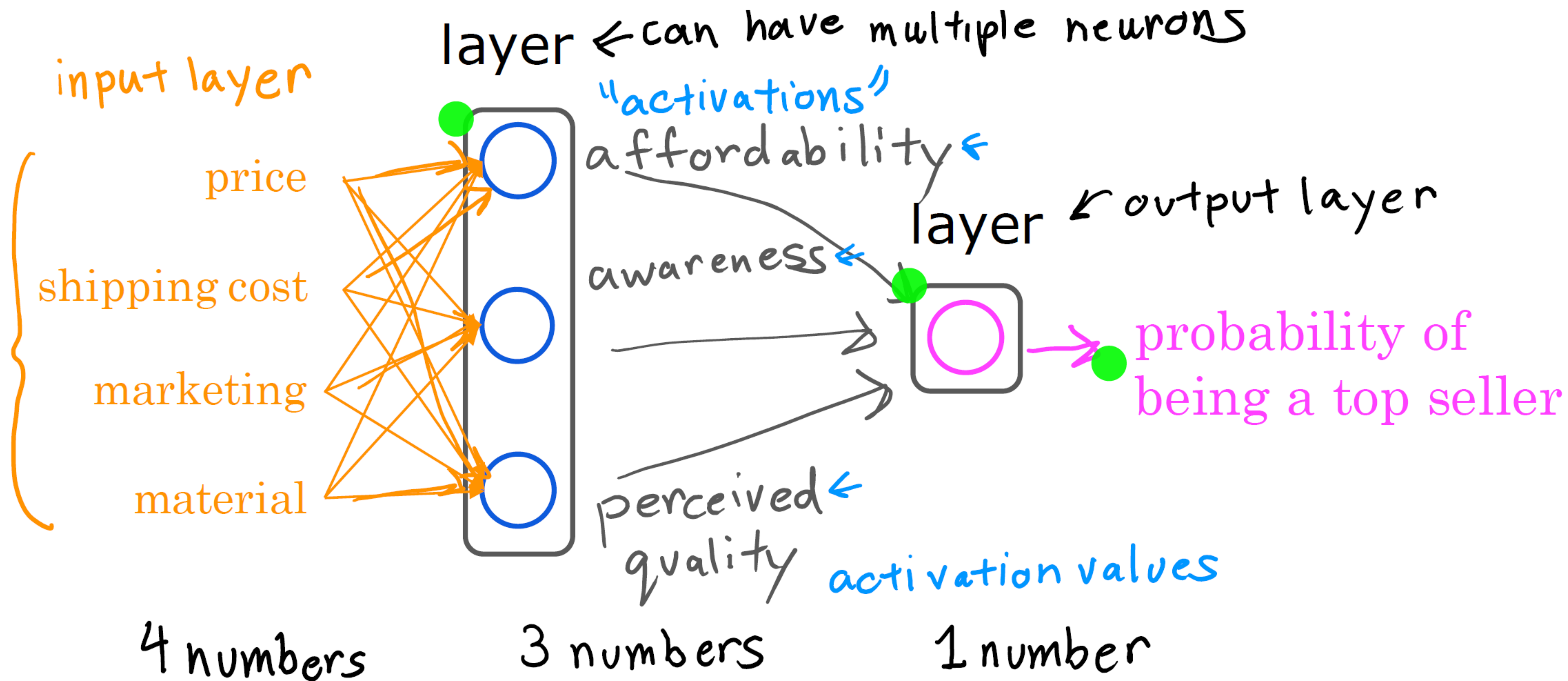
$x = \text{price}$ (input)
 $a = f(x) = \frac{1}{1 + e^{-(wx+b)}}$ (output)
 activation



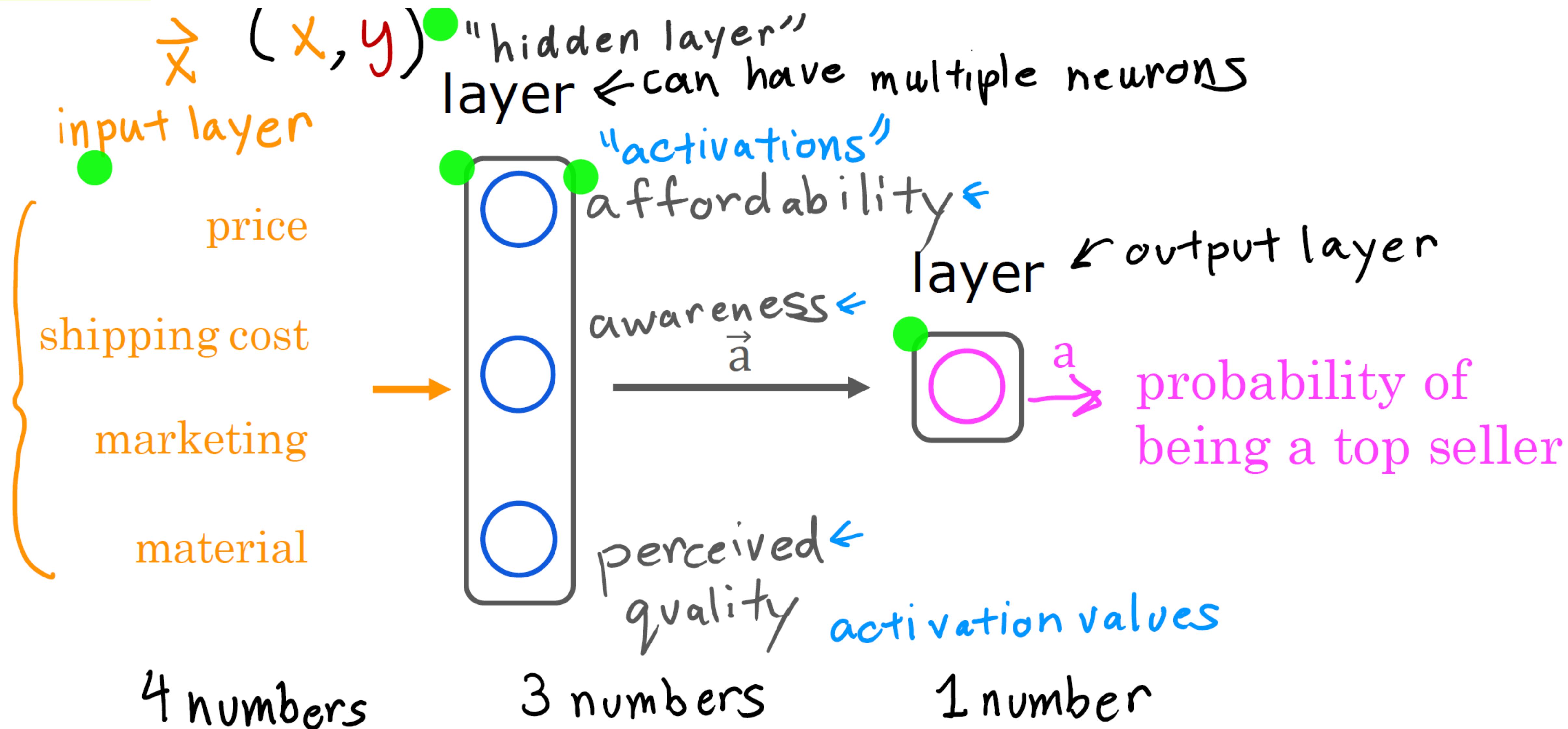
Demand Prediction



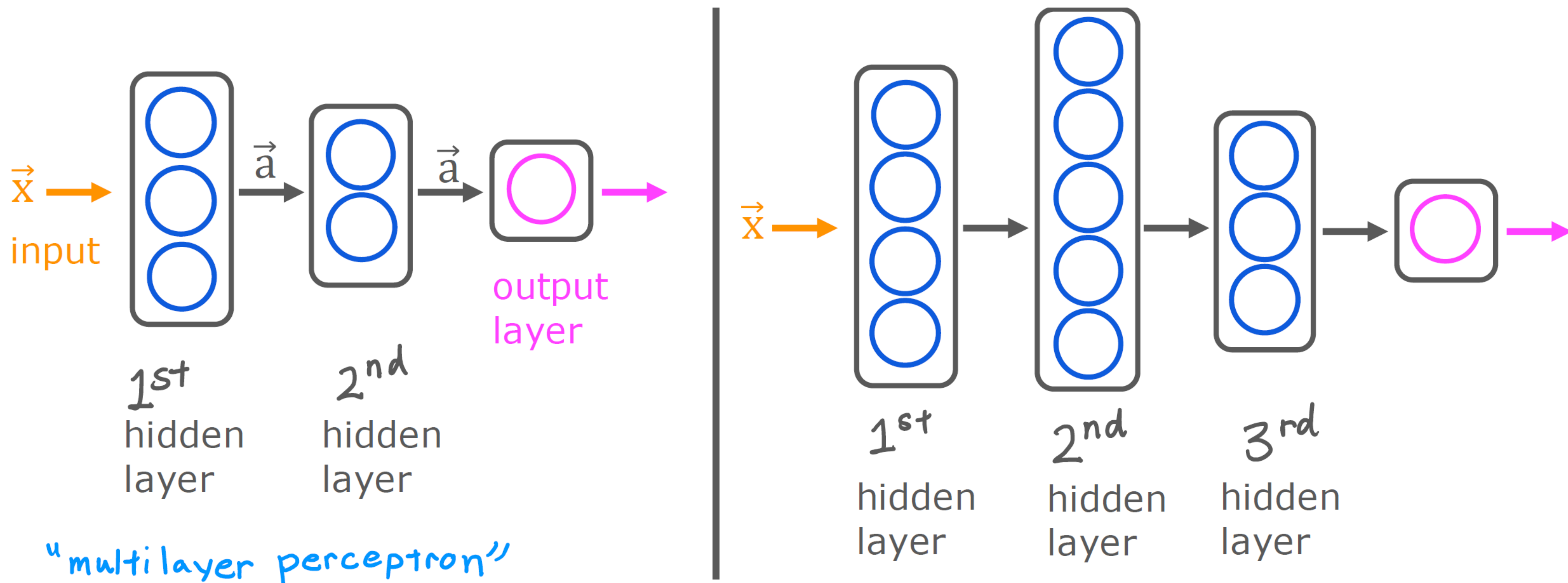
Demand Prediction



Demand Prediction

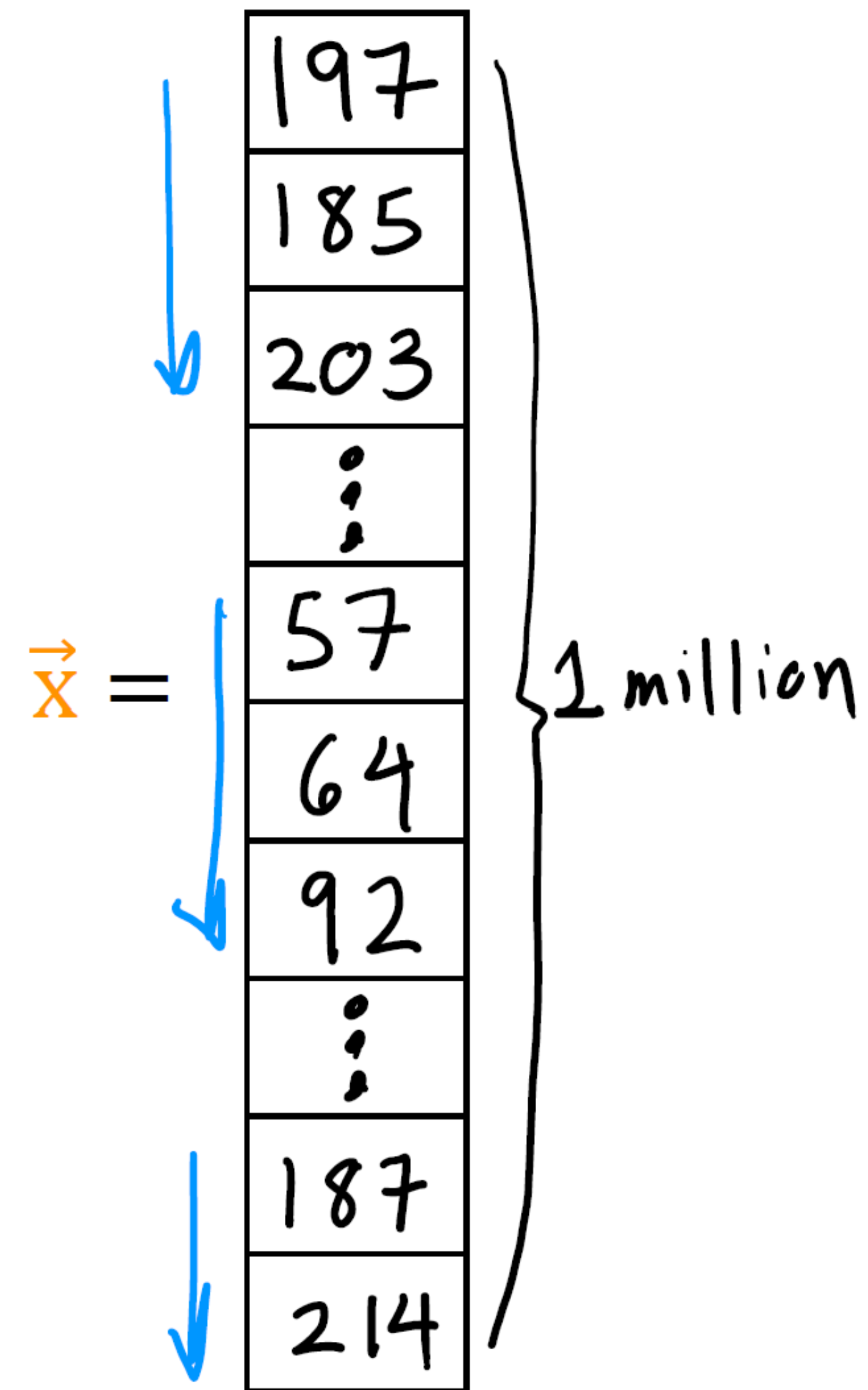
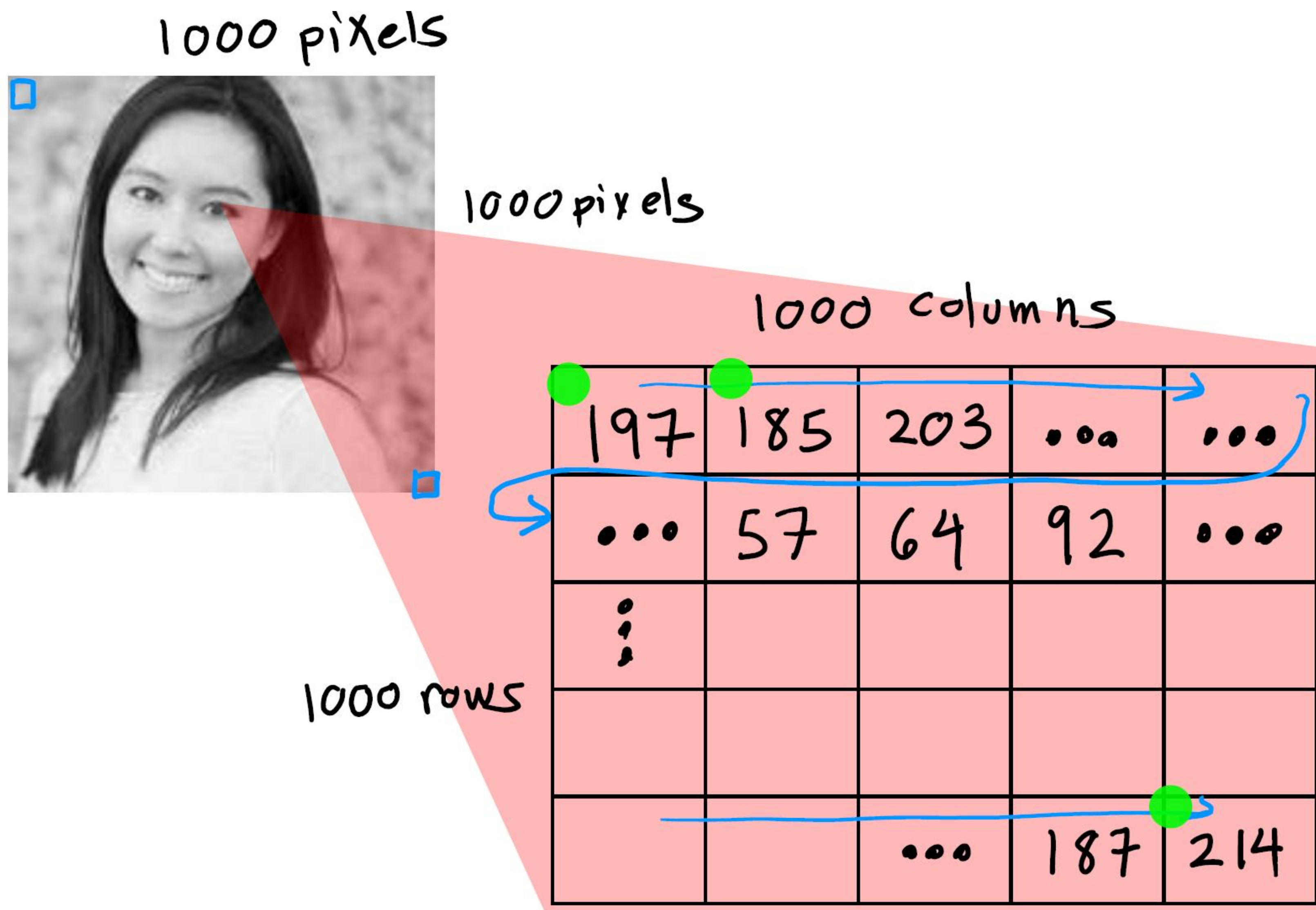


Multiple hidden layers



neural network architecture

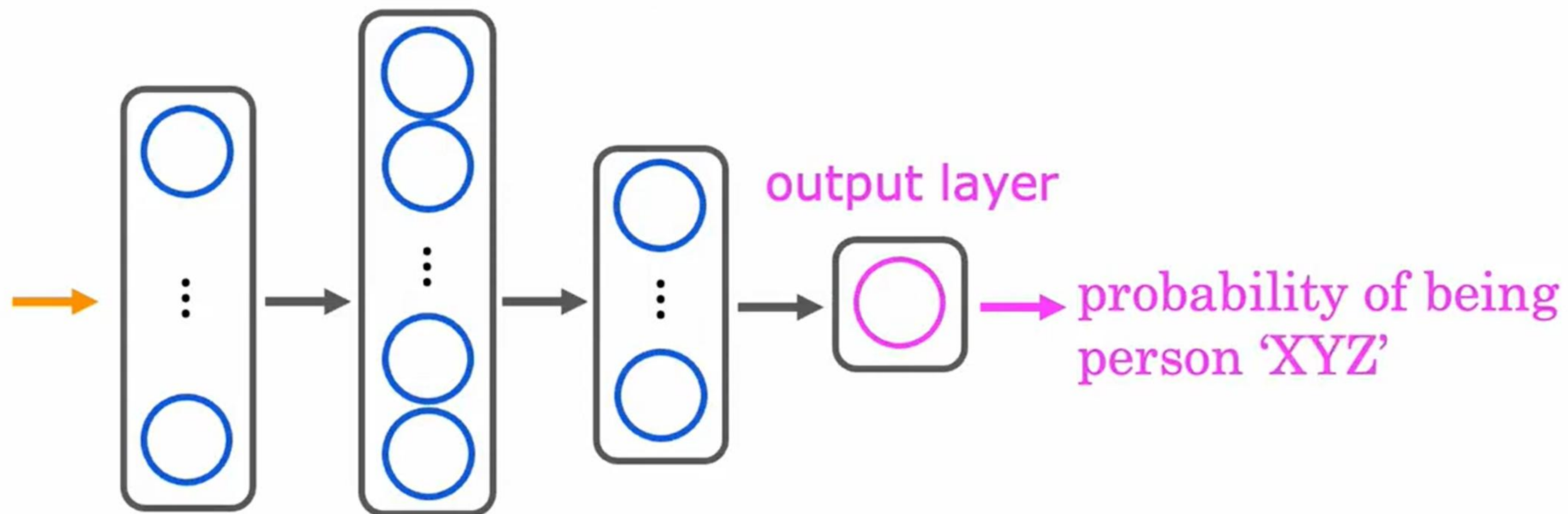
Face recognition



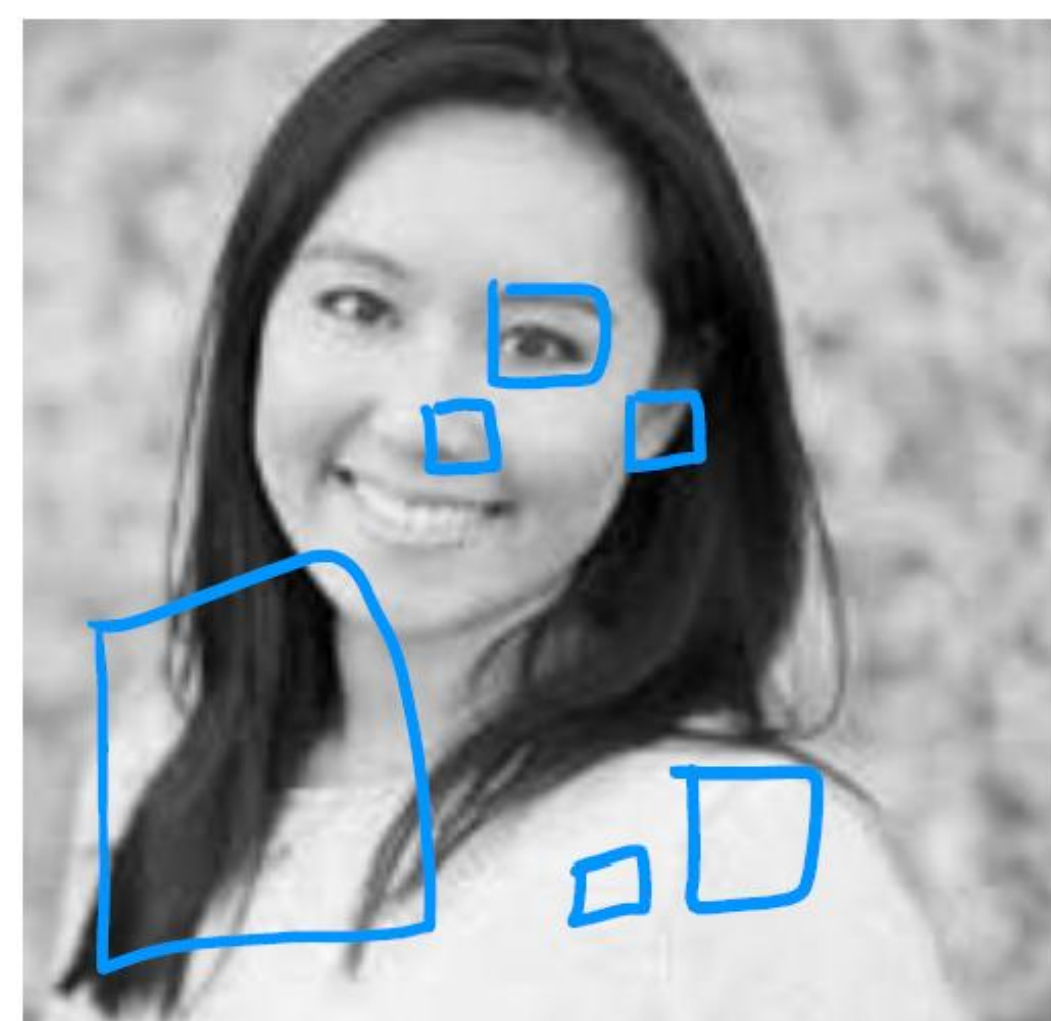
Face recognition



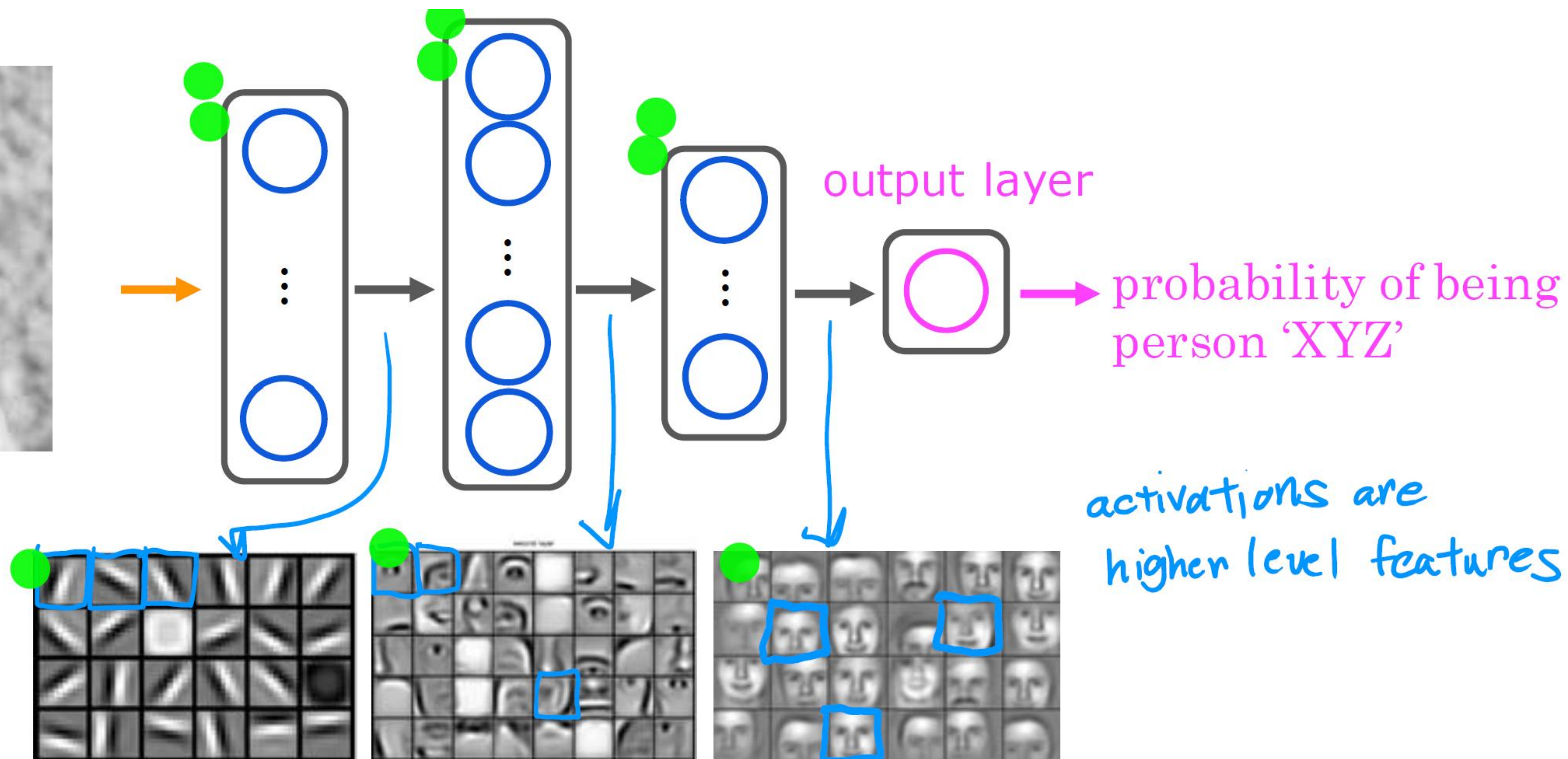
$\vec{x}$
input



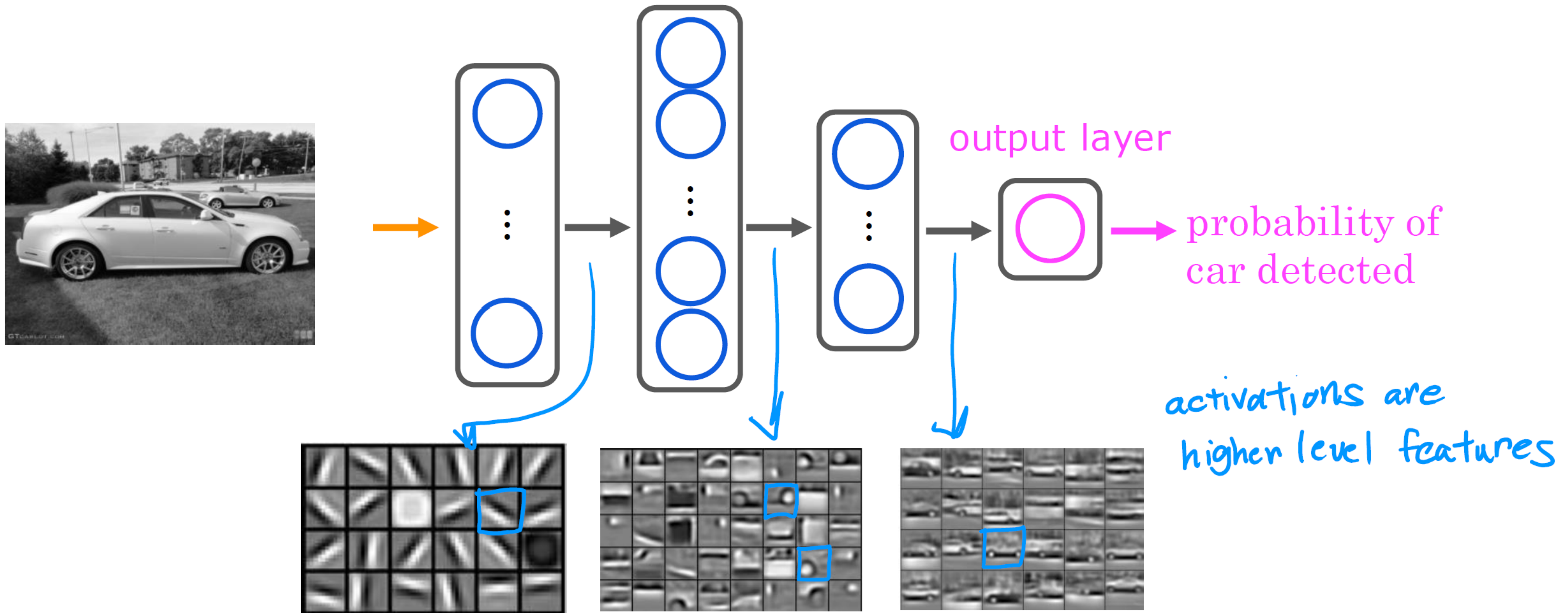
Face recognition



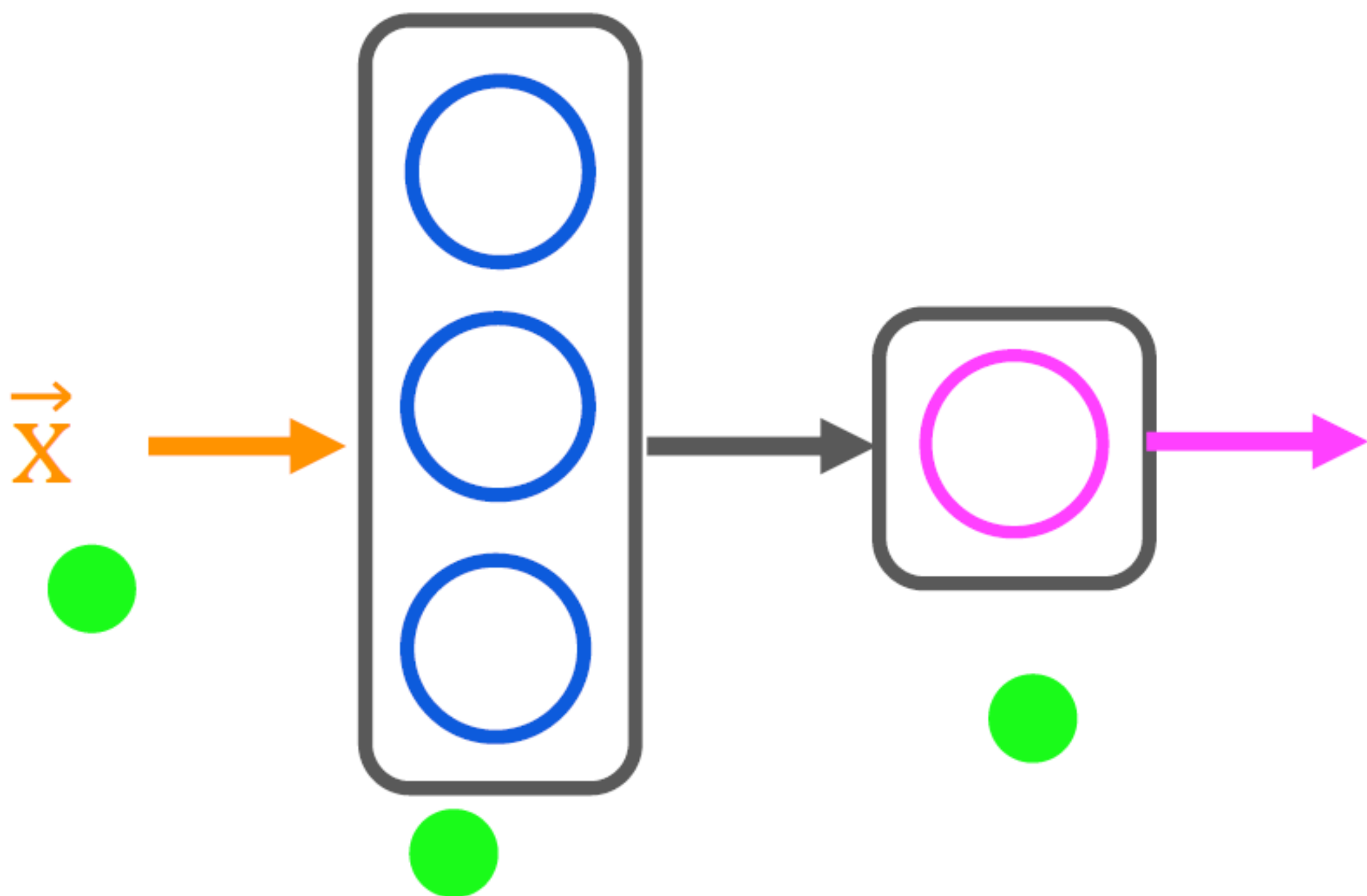
$\vec{x}$
input



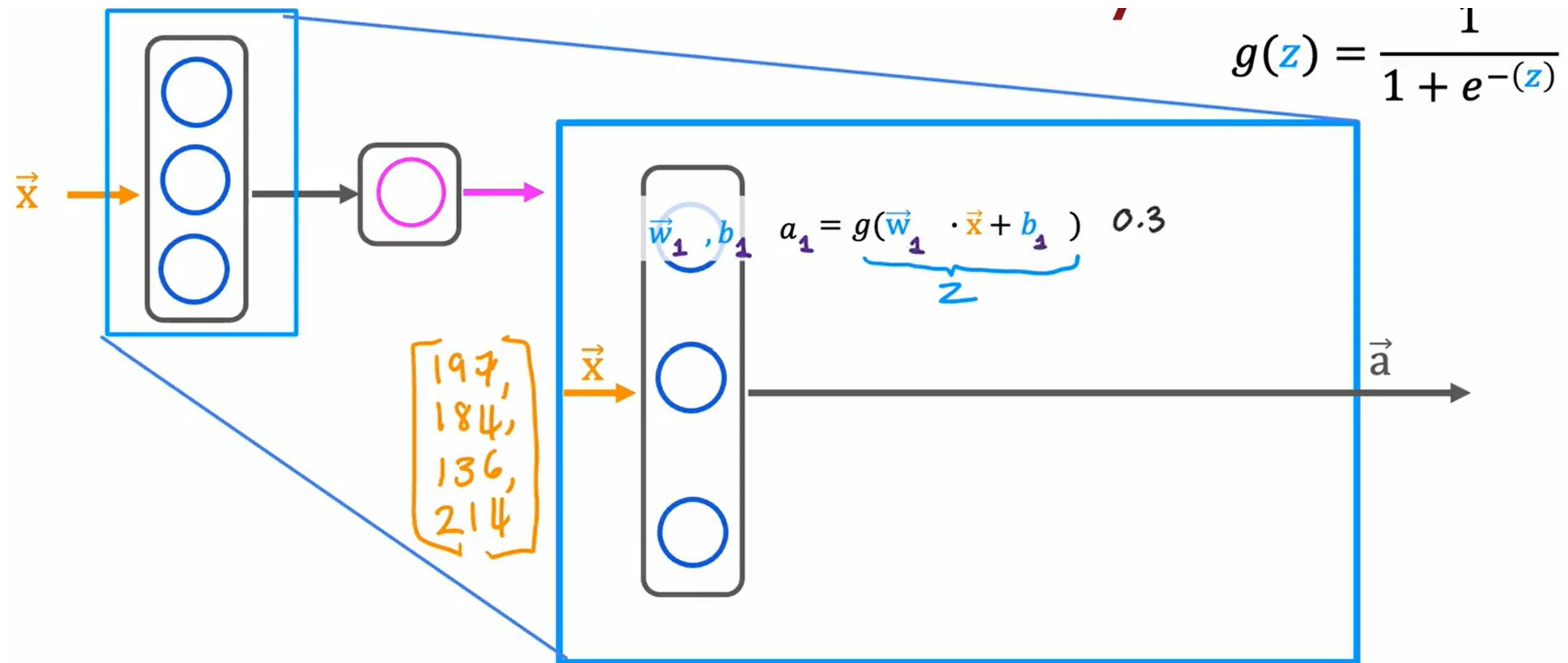
Car classification



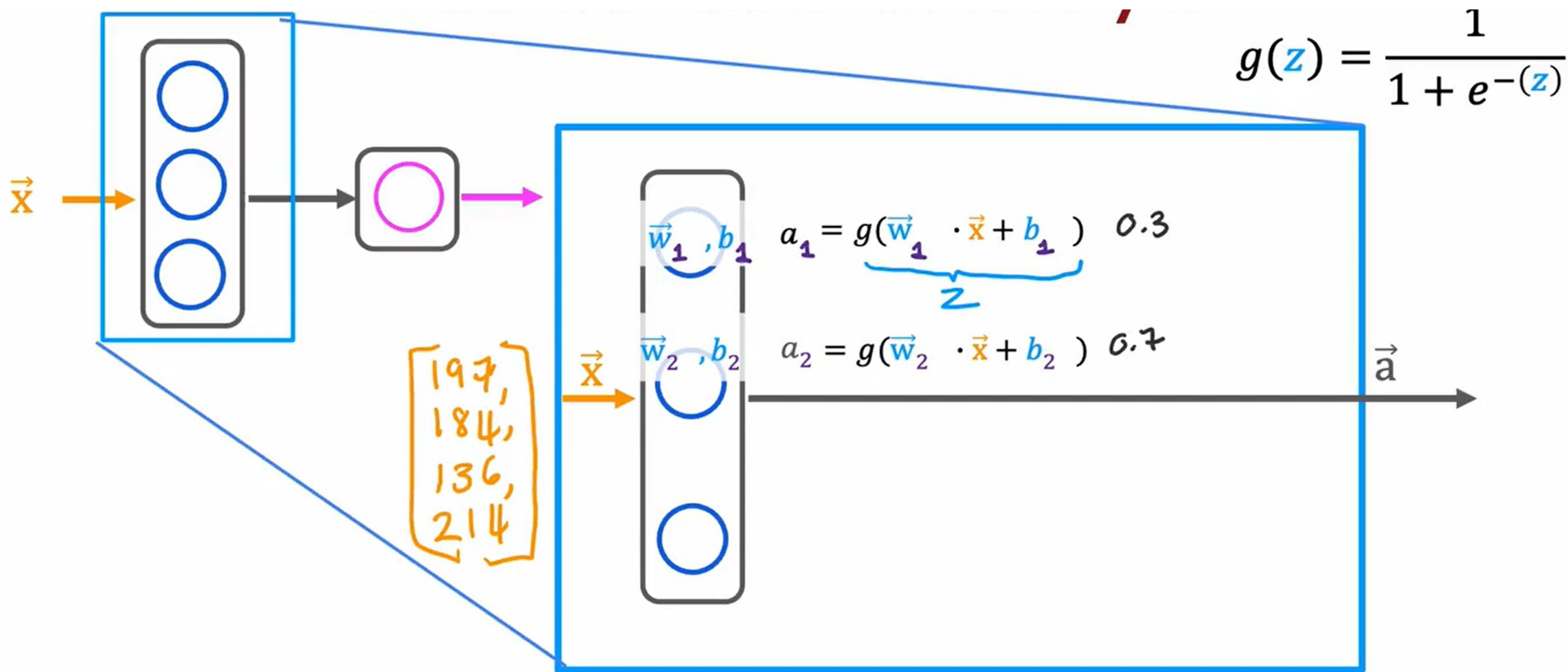
Neural network layer



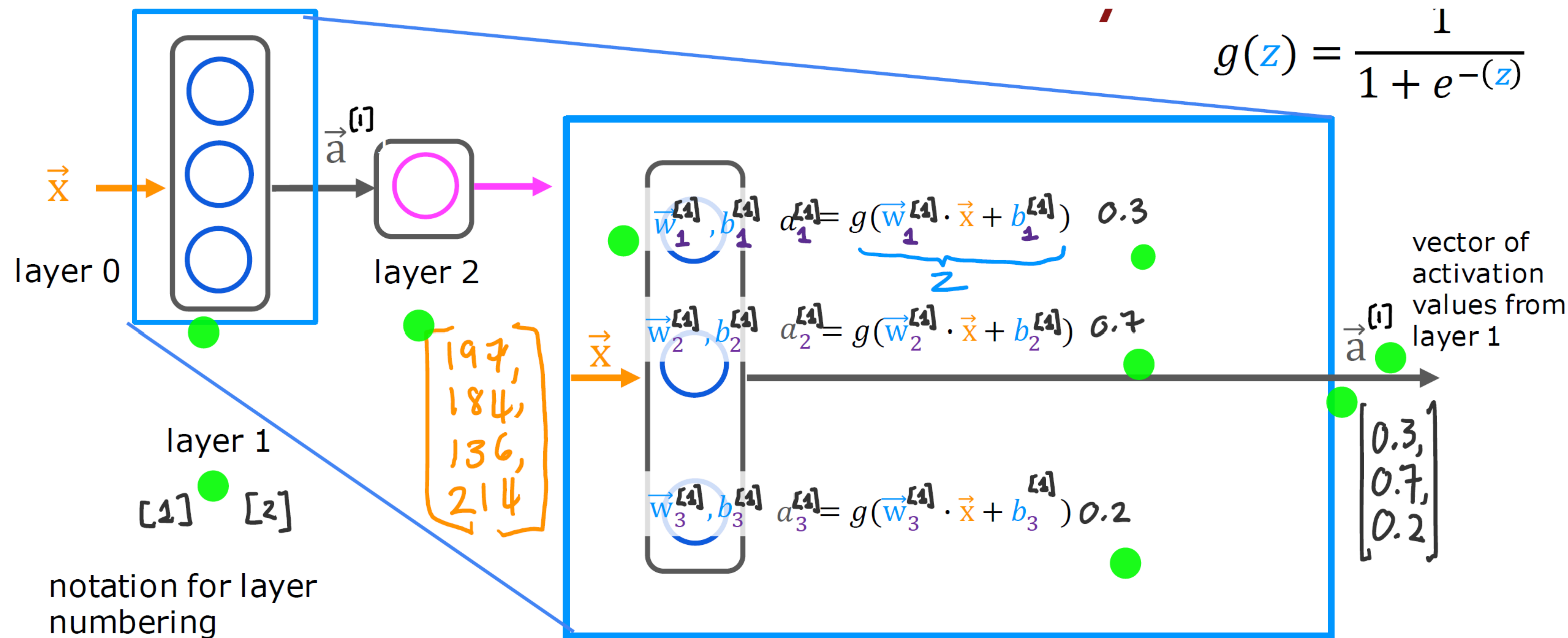
Neural network layer



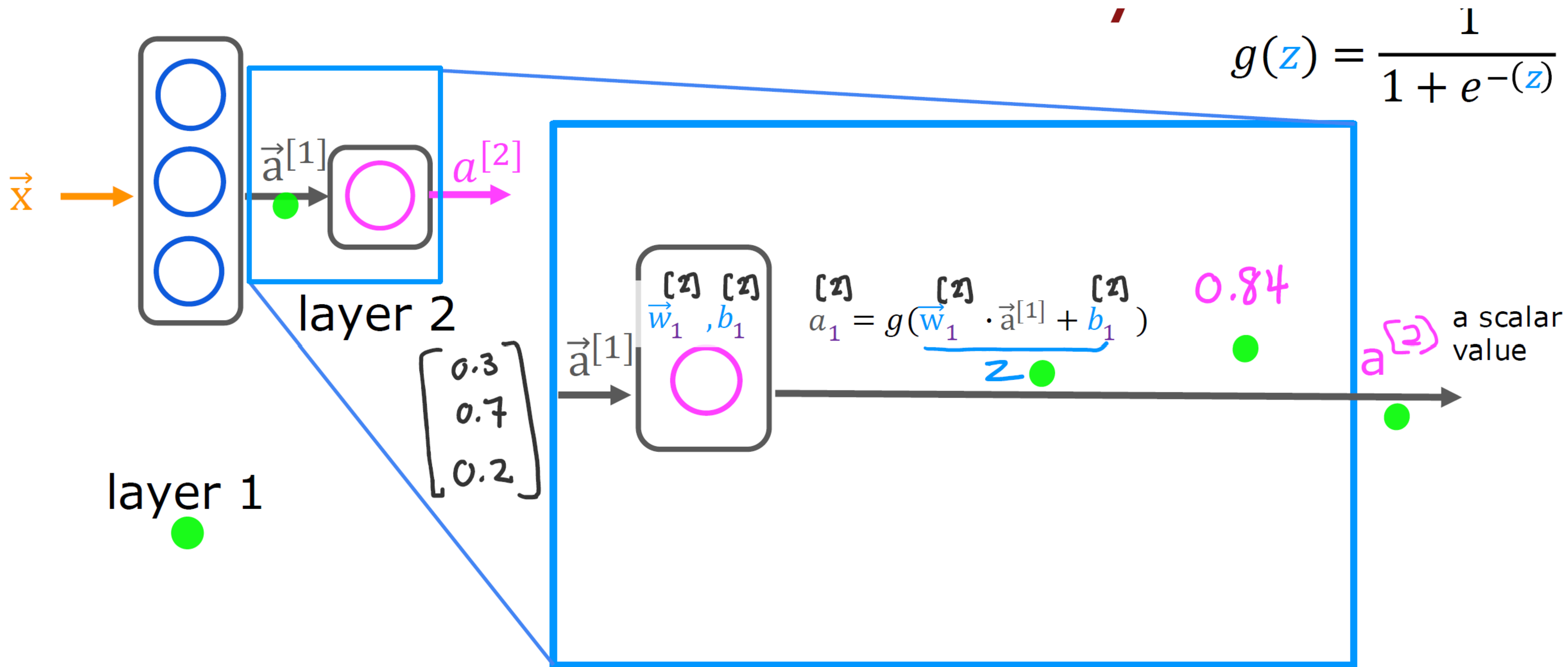
Neural network layer



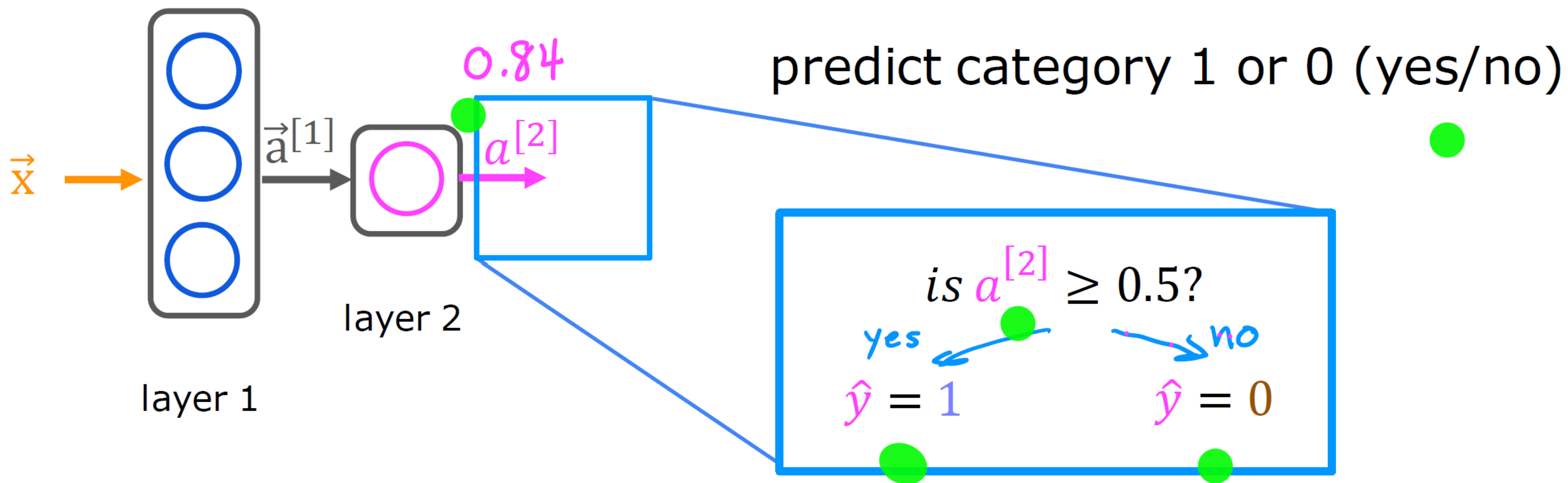
Neural network layer



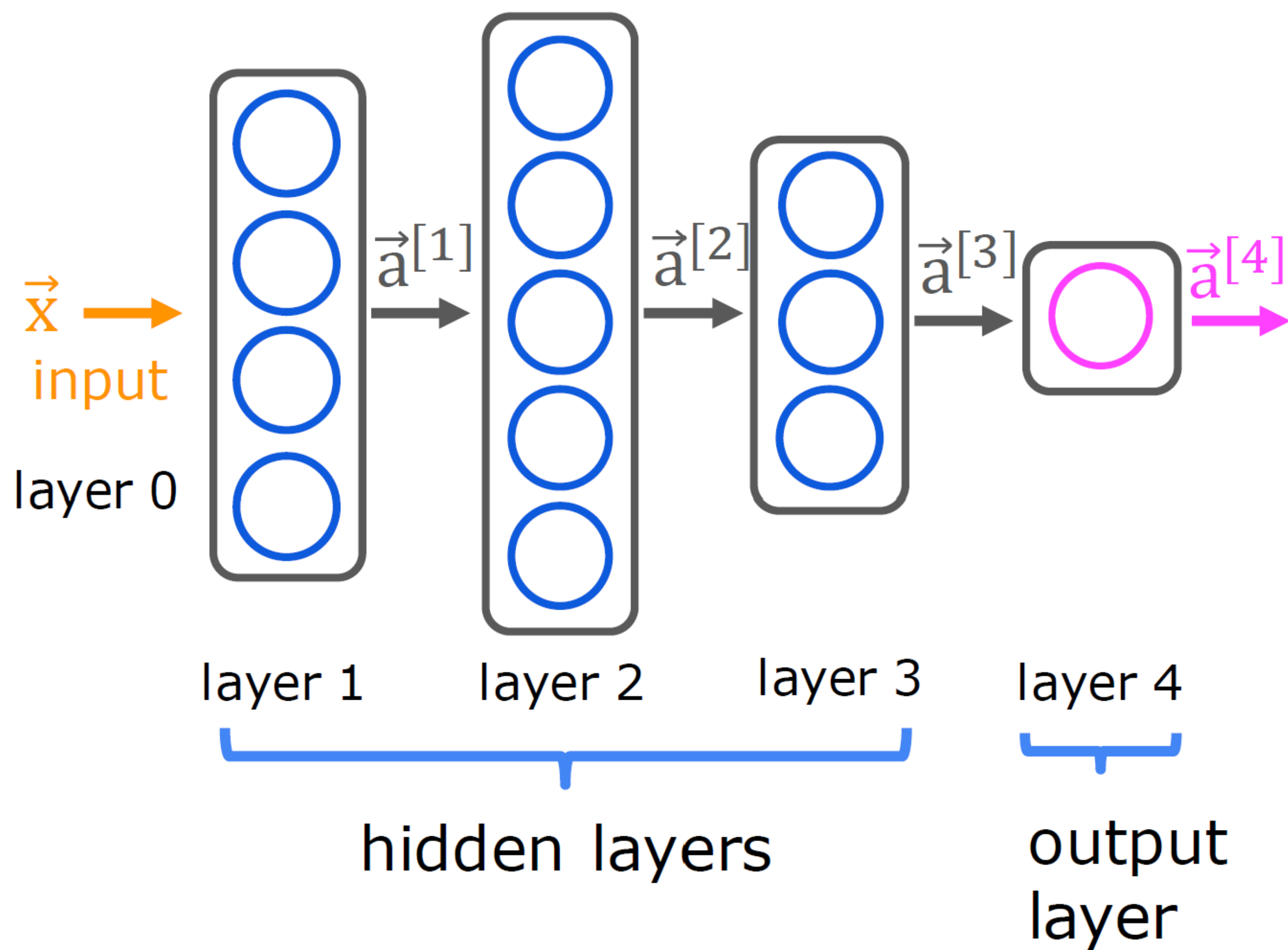
Neural network layer



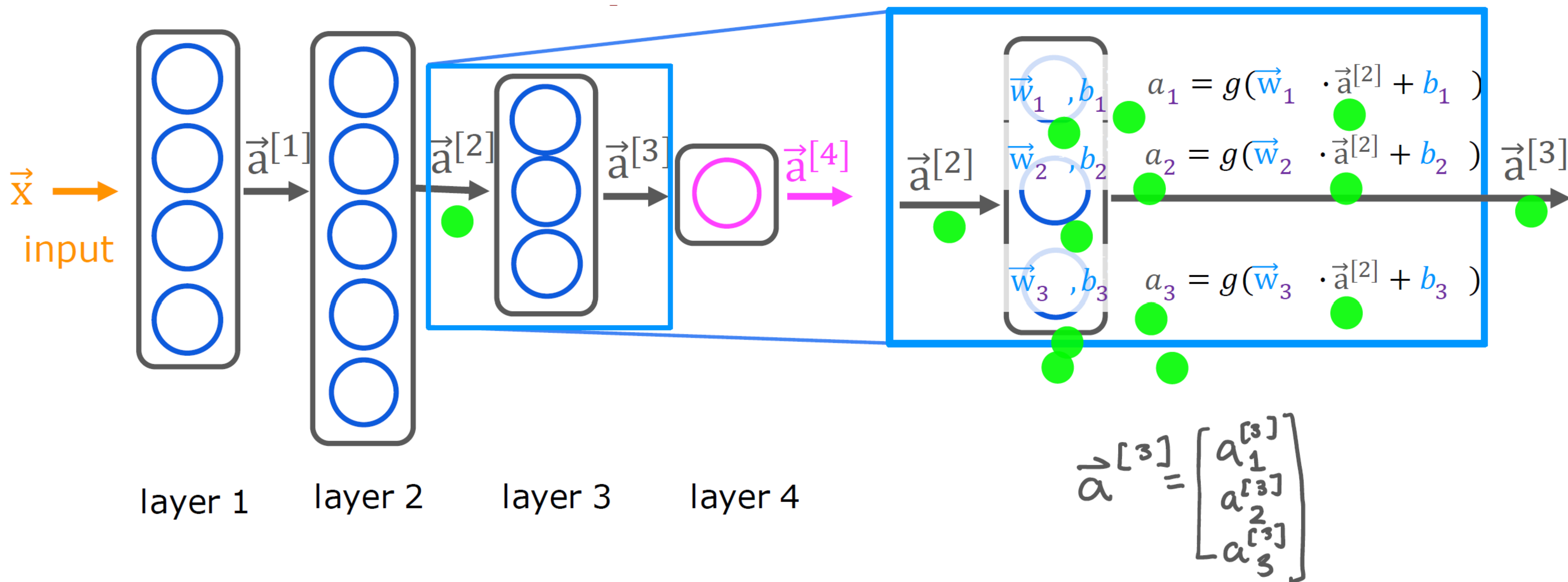
Neural network layer



More complex neural network

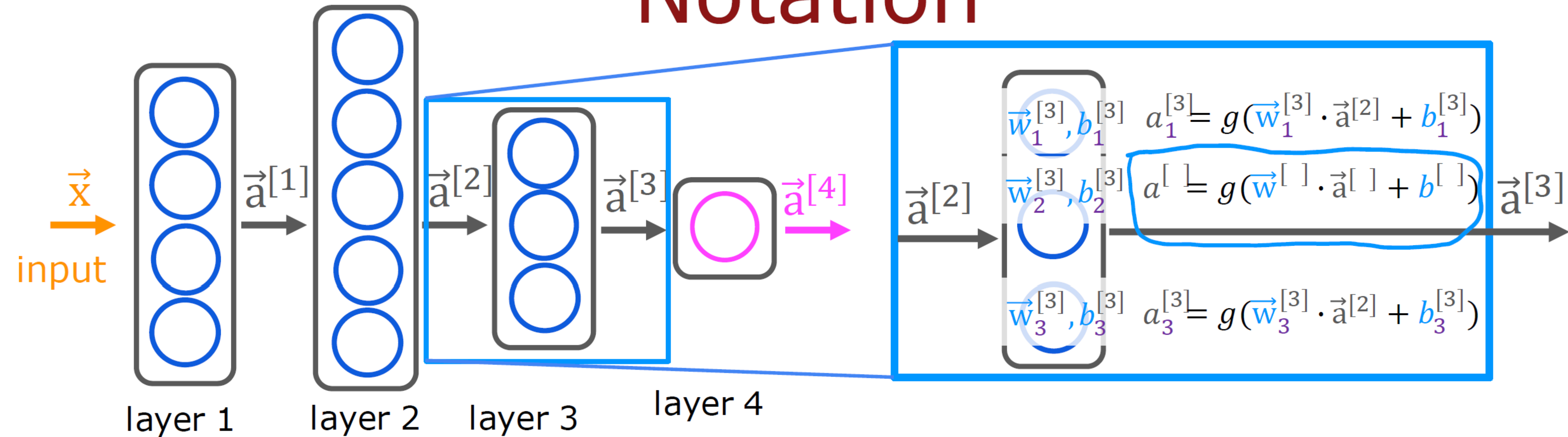


More complex neural network



More complex neural network

Notation

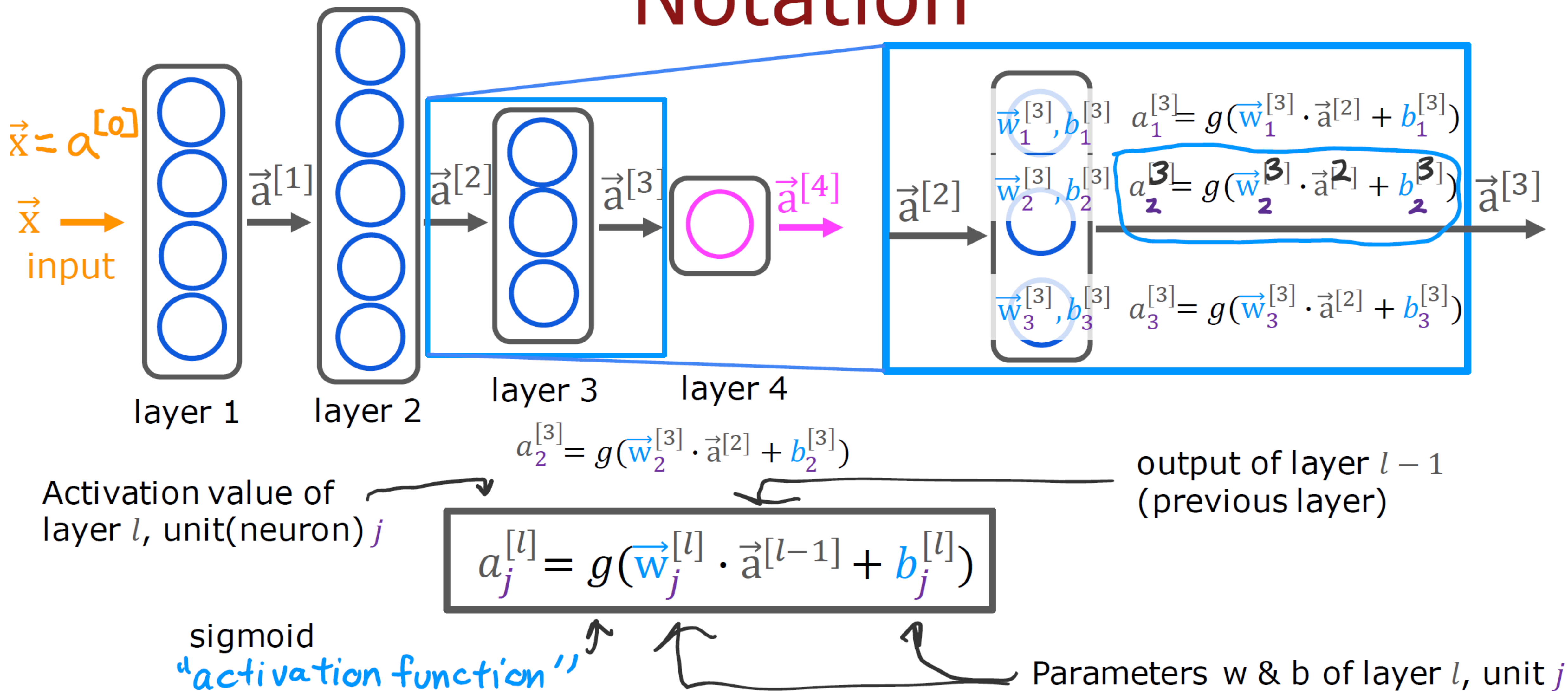


Question:

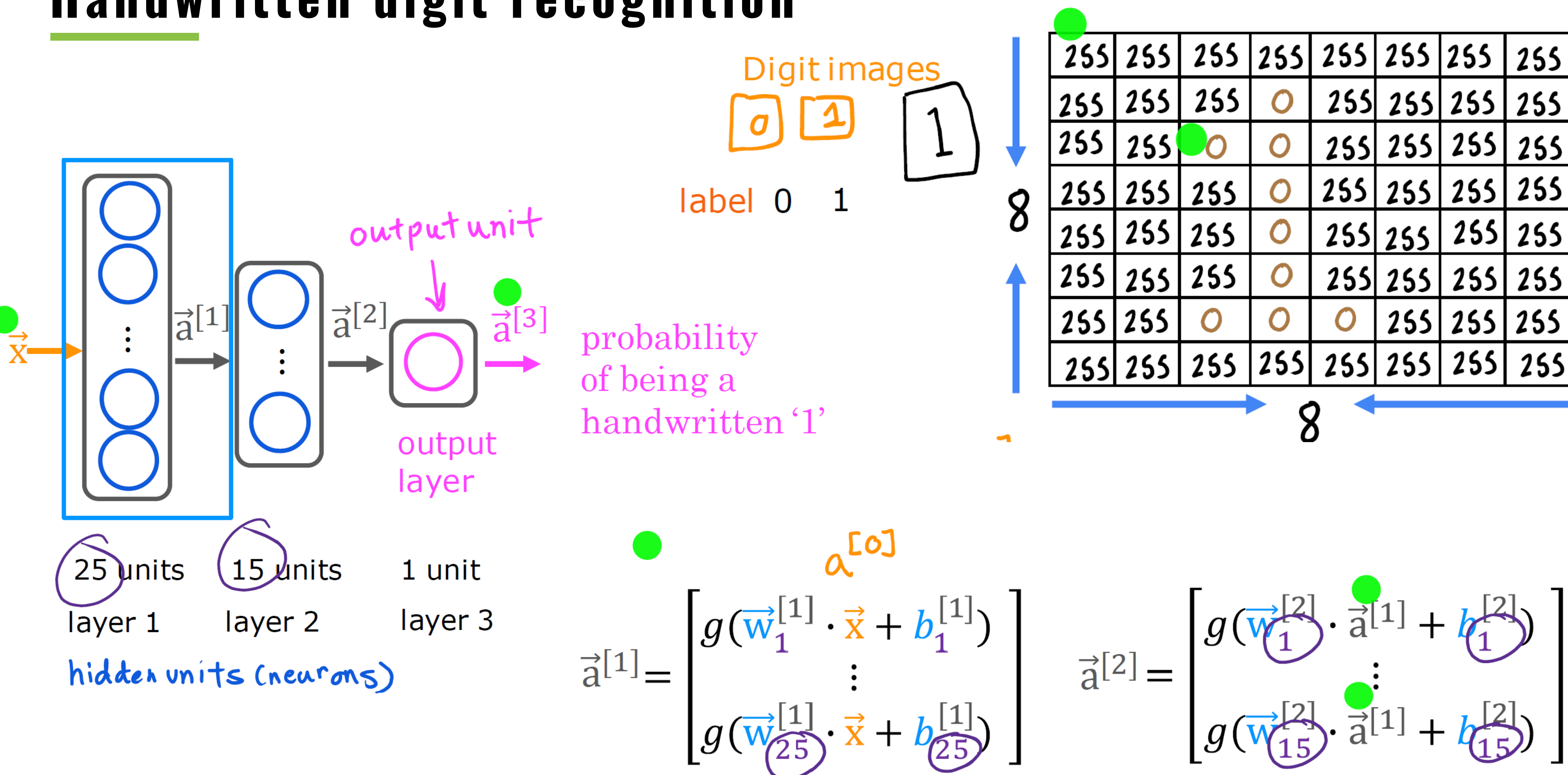
Can you fill in the superscripts and subscripts for the second neuron?

More complex neural network

Notation

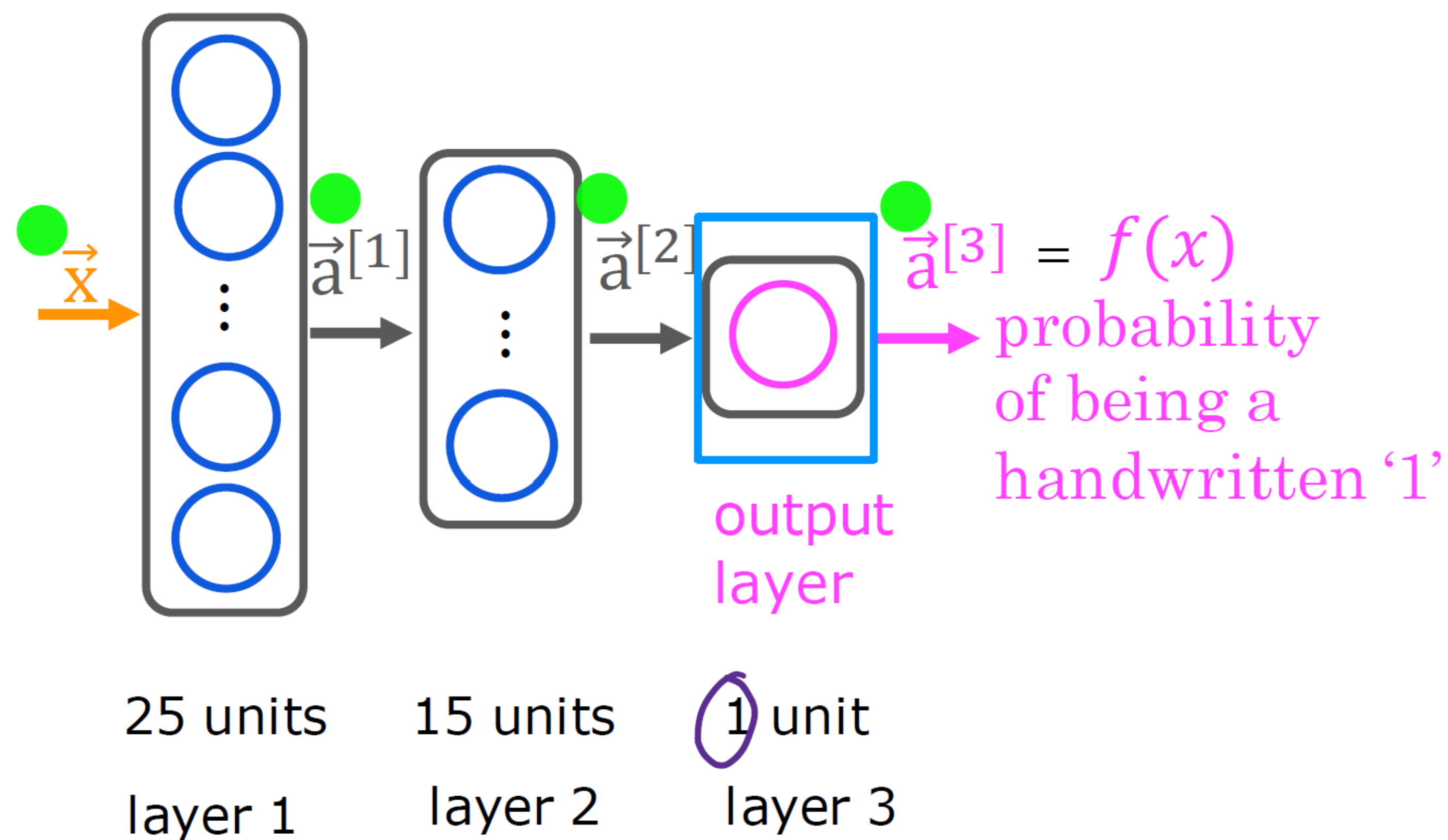


Handwritten digit recognition



Handwritten digit recognition

forward propagation



$$\vec{a}^{[3]} = \left[g \left(\vec{w}_1^{[3]} \cdot \vec{a}^{[2]} + b_1^{[3]} \right) \right]$$

is $a_1^{[3]} \geq 0.5$?

yes

no

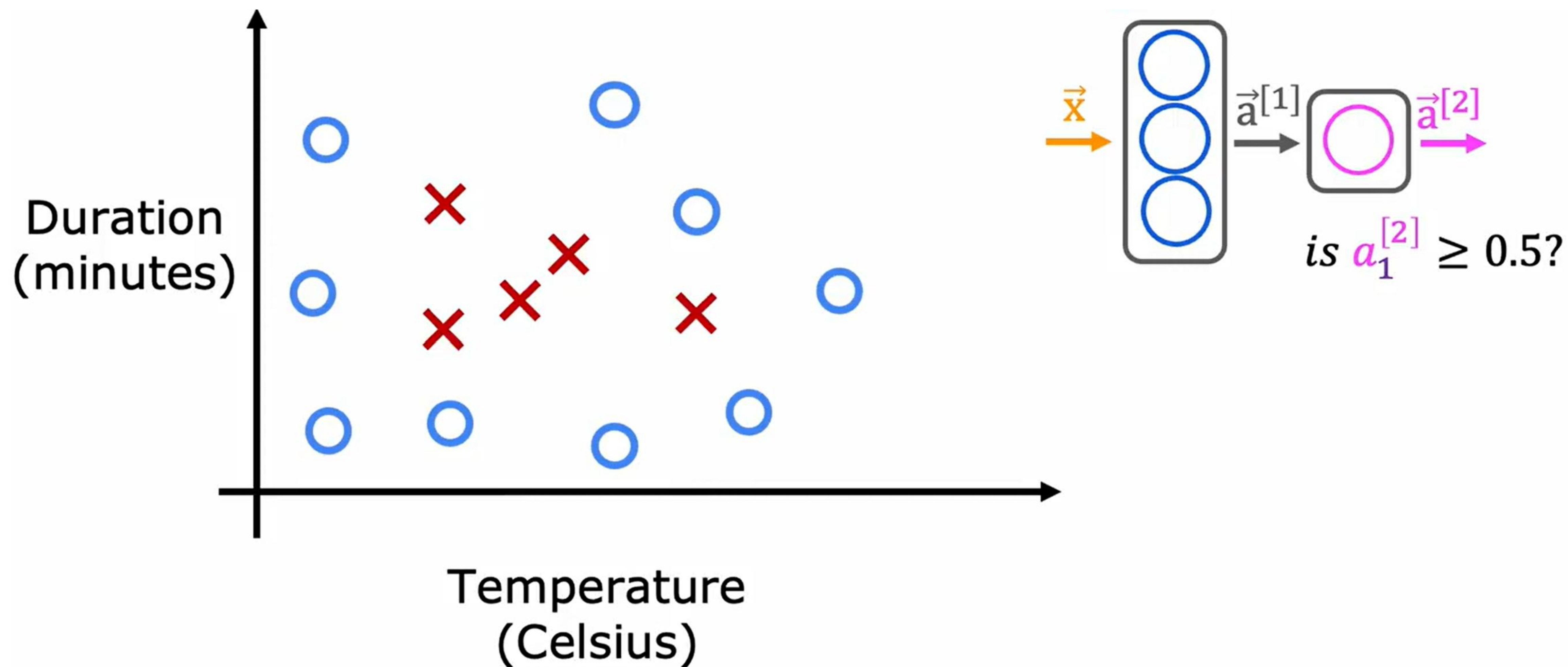
$\hat{y} = 1$

$\hat{y} = 0$

image is digit 1

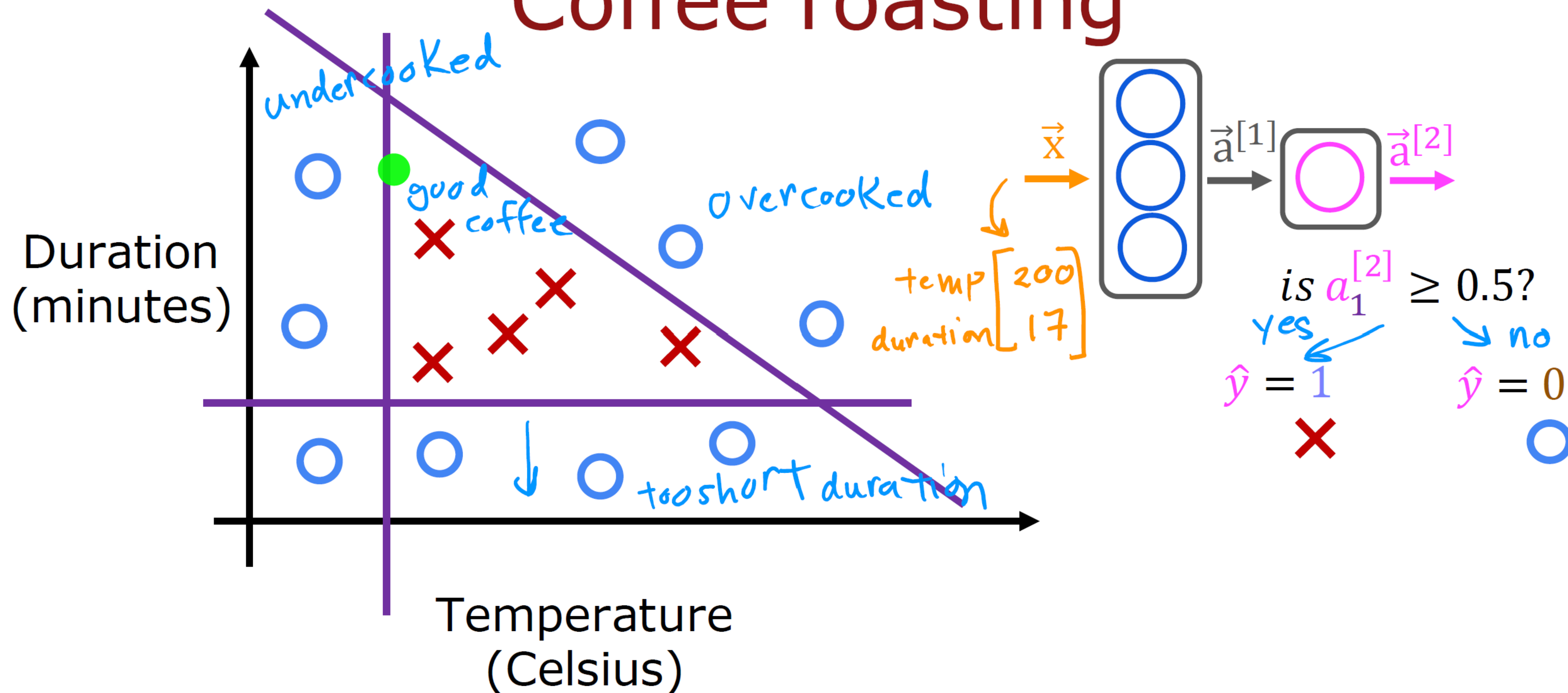
image isn't digit 1

Inference in Code

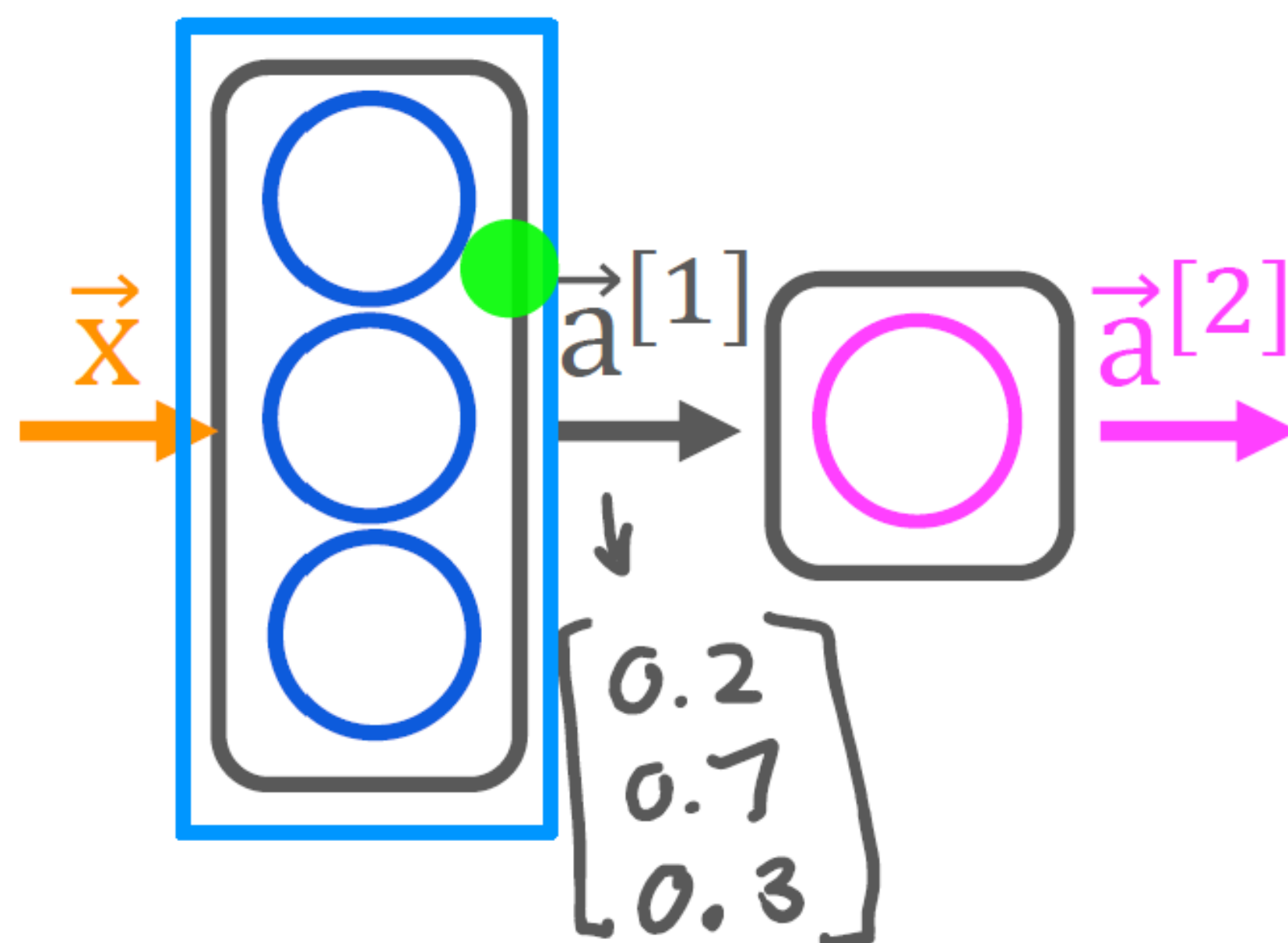


Inference in Code

Coffee roasting

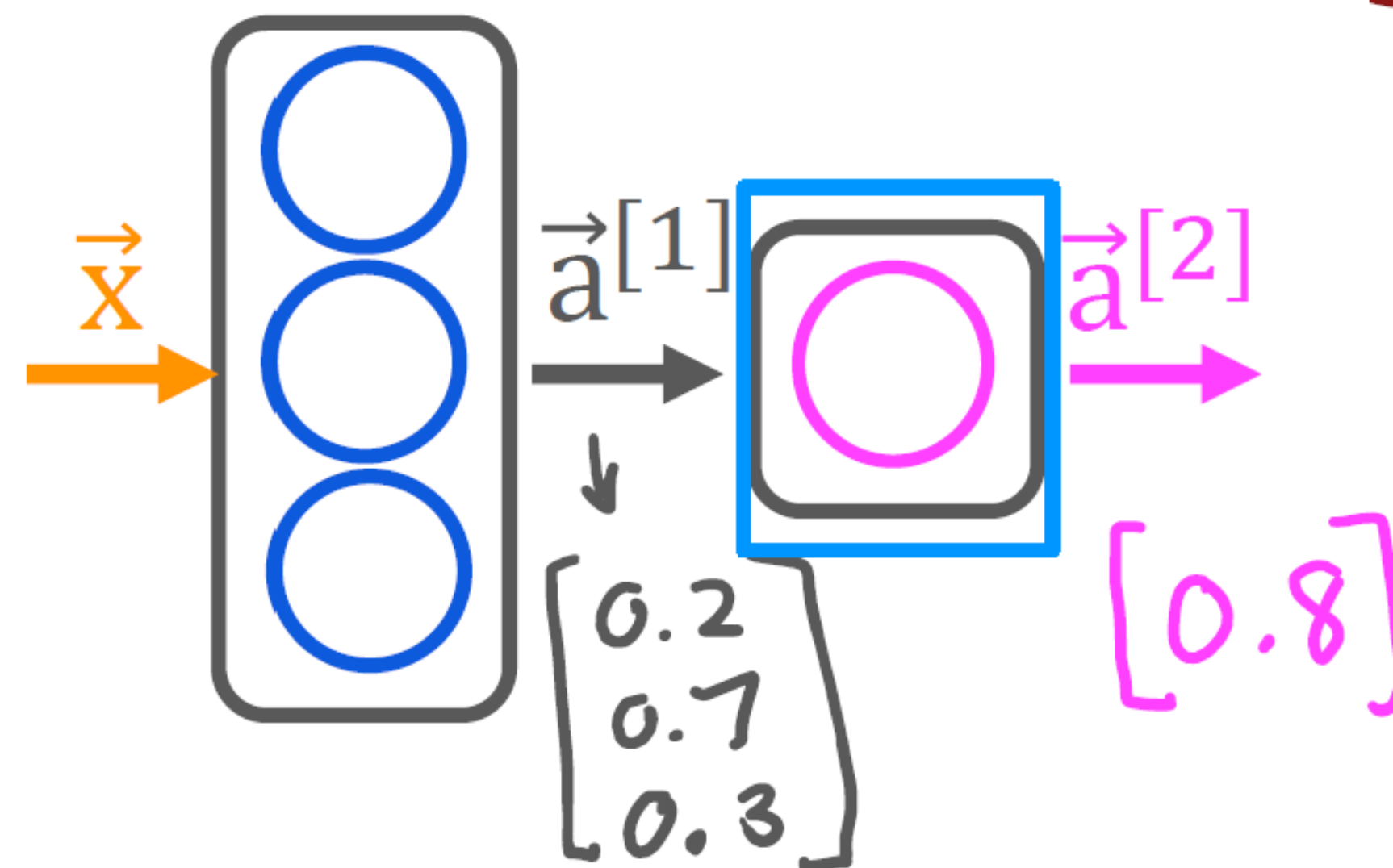


Inference in Code



```
x = np.array([[200.0, 17.0]])  
layer_1 = Dense(units=3, activation='sigmoid')  
a1 = layer_1(x)
```

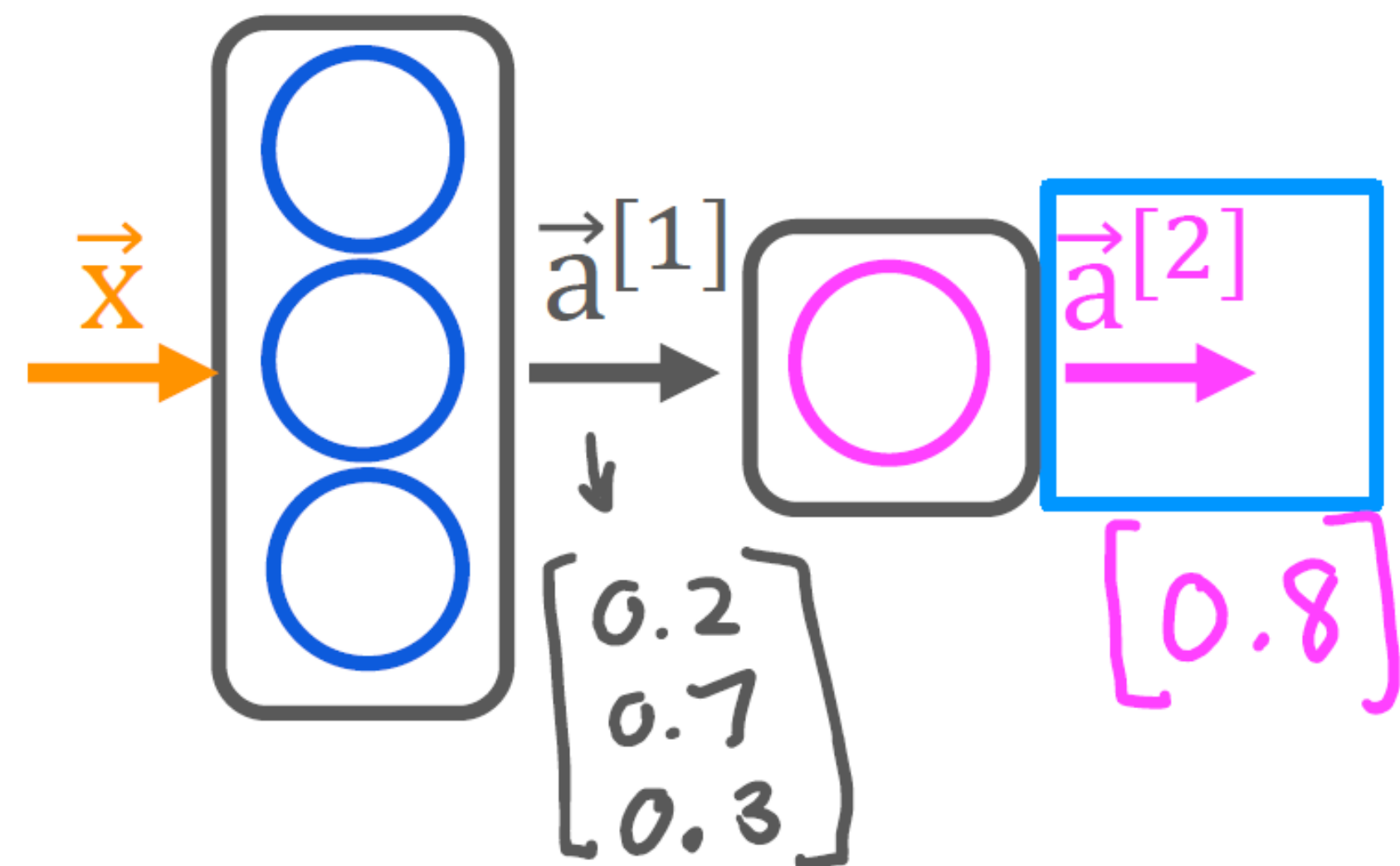
Inference in Code



```
x = np.array([[200.0, 17.0]])  
layer_1 = Dense(units=3, activation='sigmoid')  
a1 = layer_1(x)
```

```
layer_2 = Dense(units=1, activation='sigmoid')  
a2 = layer_2(a1)
```

Inference in Code



```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

is $a_1^{[2]} \geq 0.5$? ●

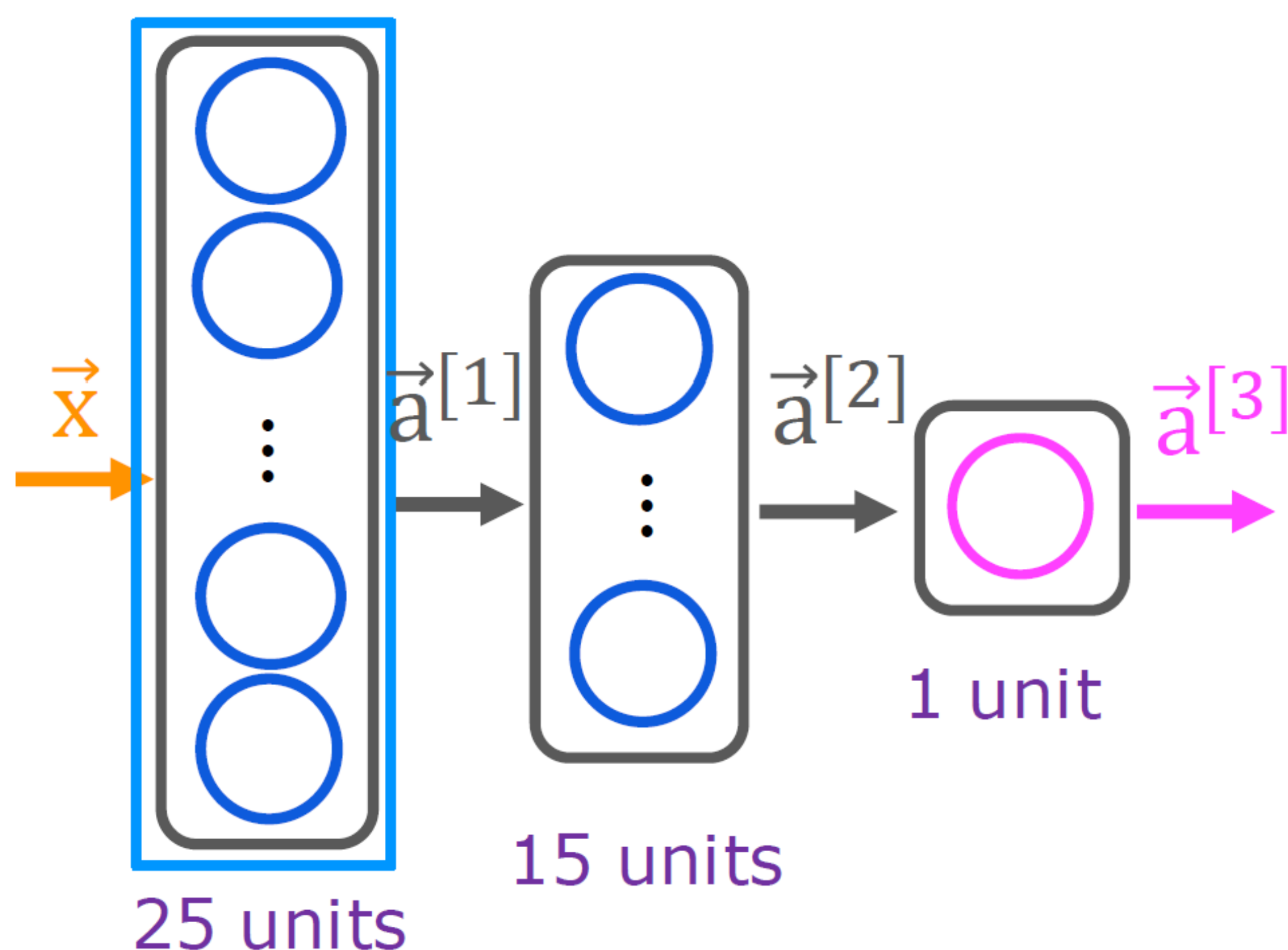
yes $\hat{y} = 1$ ✗

no $\hat{y} = 0$ ○

```
if a2 >= 0.5:
    yhat = 1
else:
    yhat = 0
```


Inference in Code

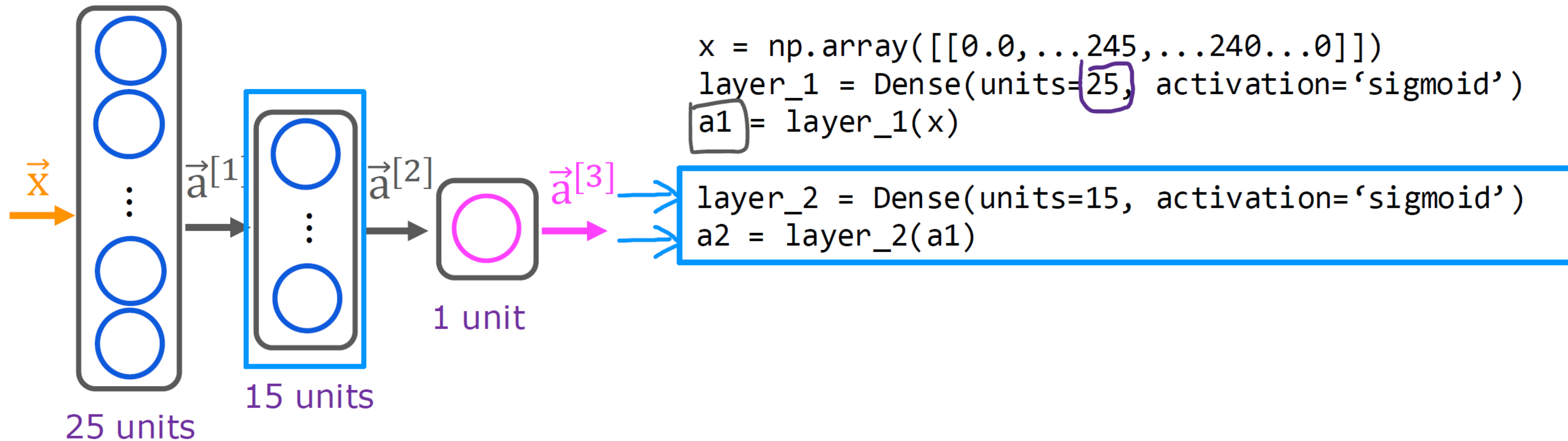
Model for digit classification



```
x = np.array([[0.0, ..., 245, ..., 240, ..., 0]])  
layer_1 = Dense(units=25, activation='sigmoid')  
a1 = layer_1(x)
```

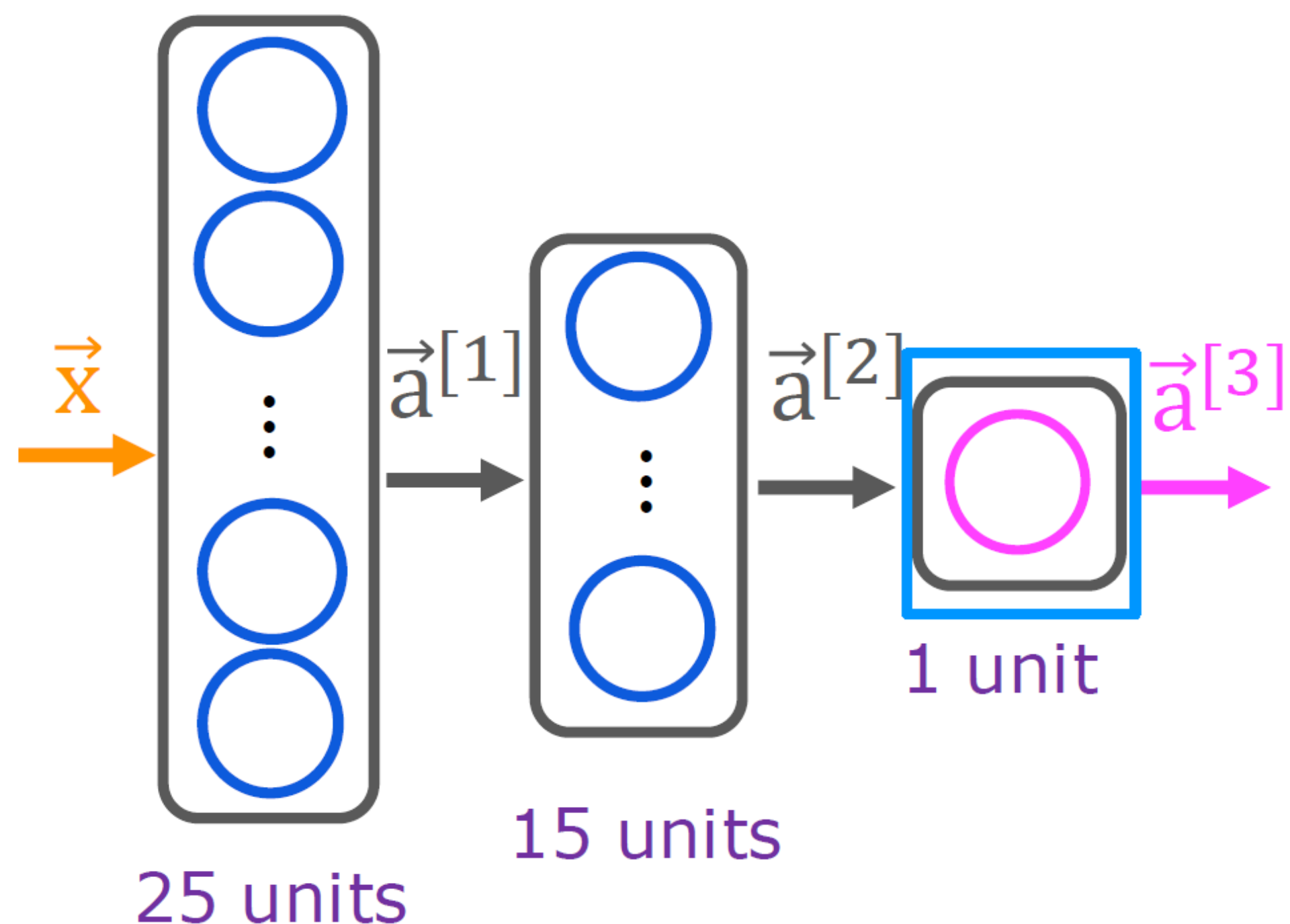
Inference in Code

Model for digit classification



Inference in Code

Model for digit classification



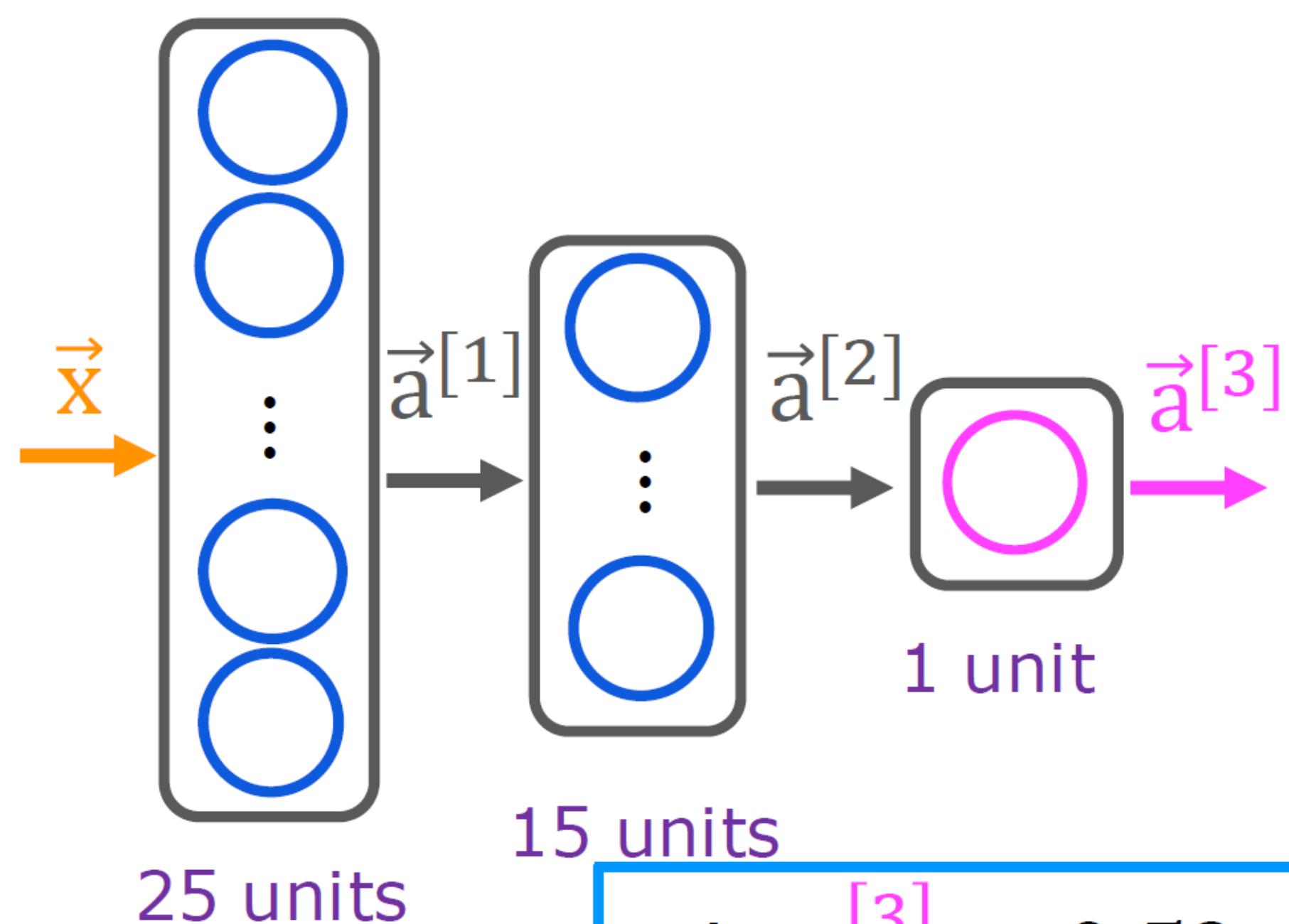
```
x = np.array([[0.0, ..., 245, ..., 240, ..., 0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)
```

```
layer_3 = Dense(units=1, activation='sigmoid')
a3 = layer_3(a2)
```


Inference in Code

Model for digit classification



```
x = np.array([[0.0, ..., 245, ..., 240, ..., 0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)
```

```
layer_3 = Dense(units=1, activation='sigmoid')
a3 = layer_3(a2)
```

is $a_1^{[3]} \geq 0.5$?

$\hat{y} = 1$

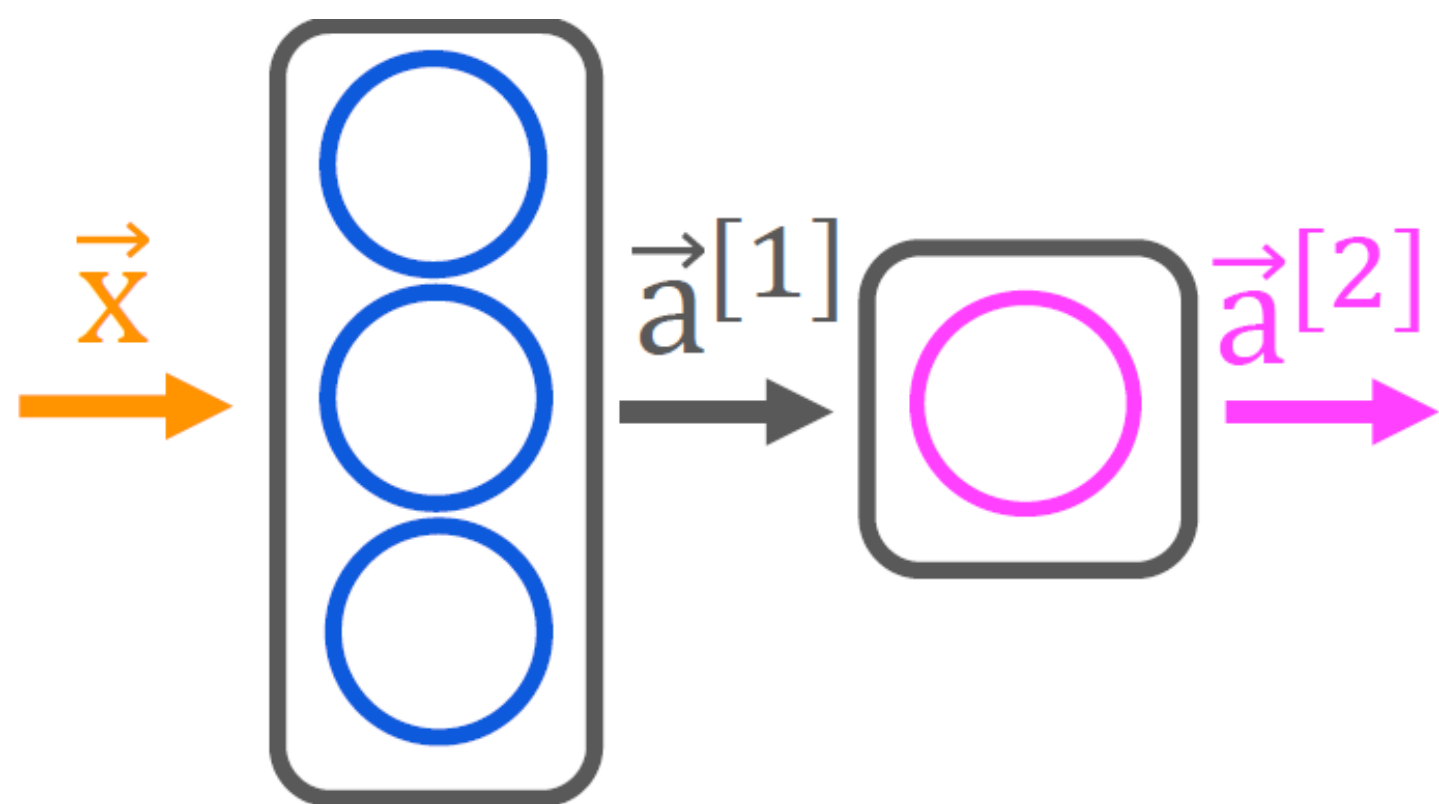
✗

$\hat{y} = 0$

○

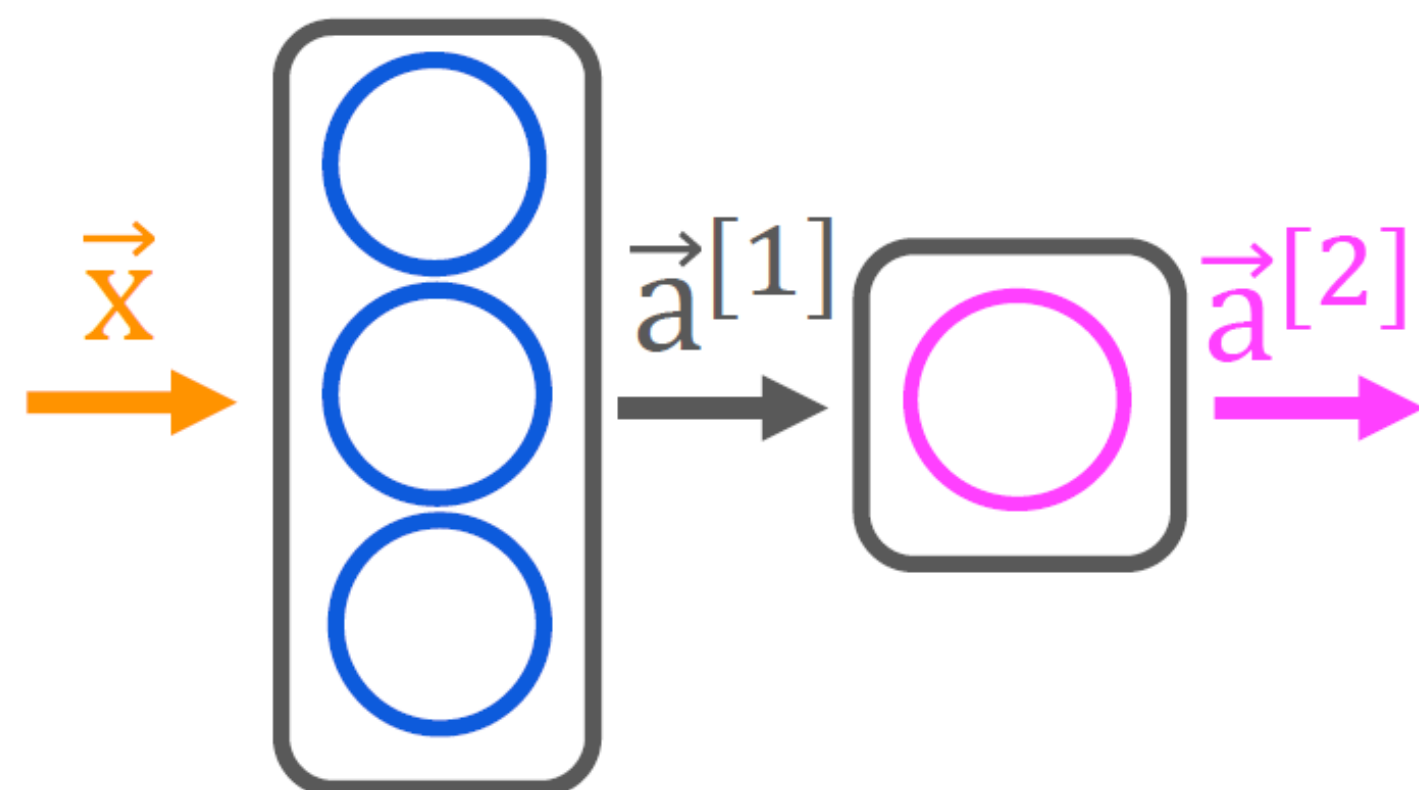
```
if a3 >= 0.5:
    yhat = 1
else:
    yhat = 0
```

Building a neural network



```
→ x = np.array([[200.0, 17.0]])  
→ layer_1 = Dense(units=3, activation="sigmoid")  
→ a1 = layer_1(x)  
  
→ layer_2 = Dense(units=1, activation="sigmoid")  
→ a2 = layer_2(a1)
```

Building a neural network



```
→ model = Sequential([
→   Dense(units=3, activation="sigmoid"),
→   Dense(units=1, activation="sigmoid")])
```

			y
●	200	17	1
	120	5	0
	425	20	0
	212	18	1

targets

```
x = np.array([[200.0, 17.0],
               [120.0, 5.0],
               [425.0, 20.0],
               [212.0, 18.0]])
```

4 x 2

```
y = np.array([1, 0, 0, 1])
```

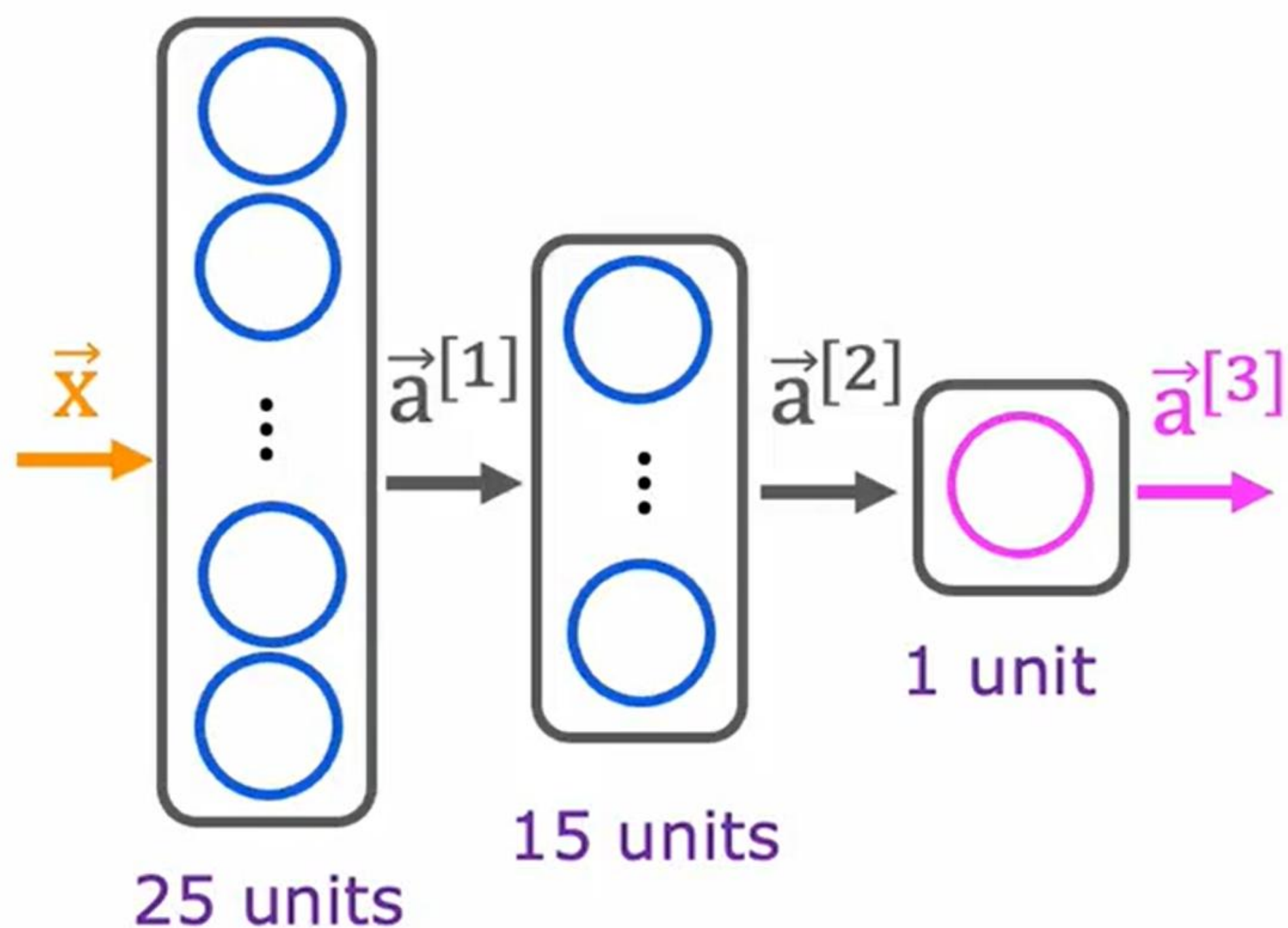
```
model.compile(...)
```

```
model.fit(x, y)
```

```
→ model.predict(x_new) ←
```

← more about this next week!

Digit classification model



```
layer_1 = Dense(units=25, activation="sigmoid")
layer_2 = Dense(units=15, activation="sigmoid")
layer_3 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2, layer_3])
```

```
model.compile(...)
```

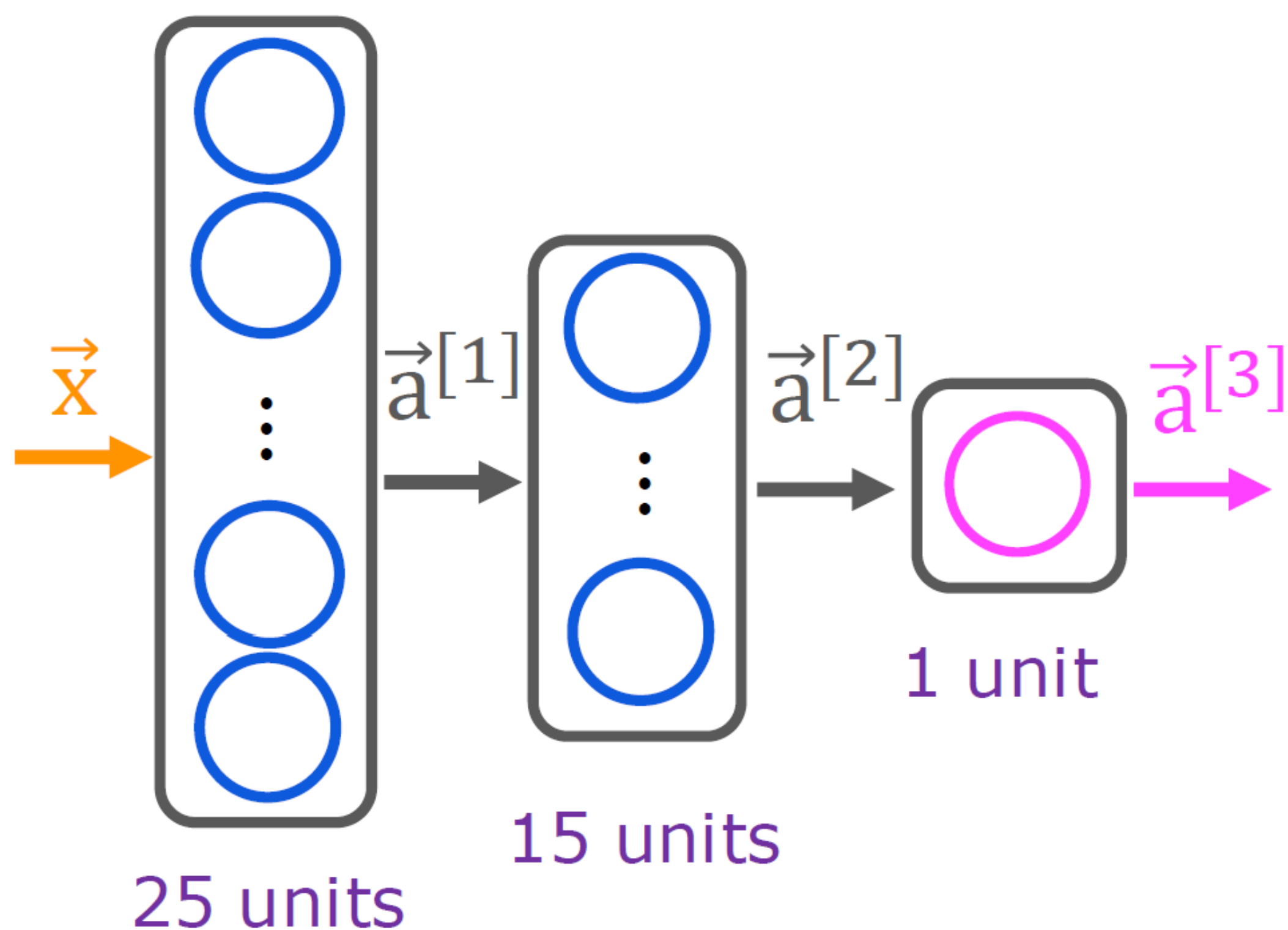
```
x = np.array([[0..., 245, ..., 17],
              [0..., 200, ..., 184]])
```

```
y = np.array([1,0])
```

```
model.fit(x,y)
```

```
model.predict(x_new)
```

Digit classification model



```

● model = Sequential([
    → Dense(units=25, activation="sigmoid"),
    → Dense(units=15, activation="sigmoid"),
    → Dense(units=1, activation="sigmoid")])

model.compile(...)

x = np.array([[0..., 245, ..., 17],
              [0..., 200, ..., 184]])
y = np.array([1, 0])

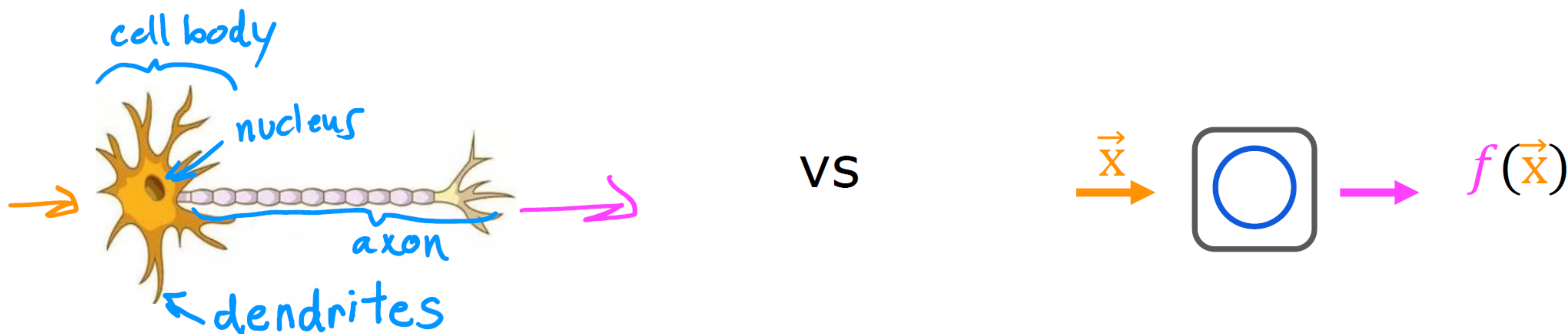
model.fit(x, y)

model.predict(x_new)

```

Neural network and the brain

Can we mimic the human brain?



We have (almost) no idea how the brain works

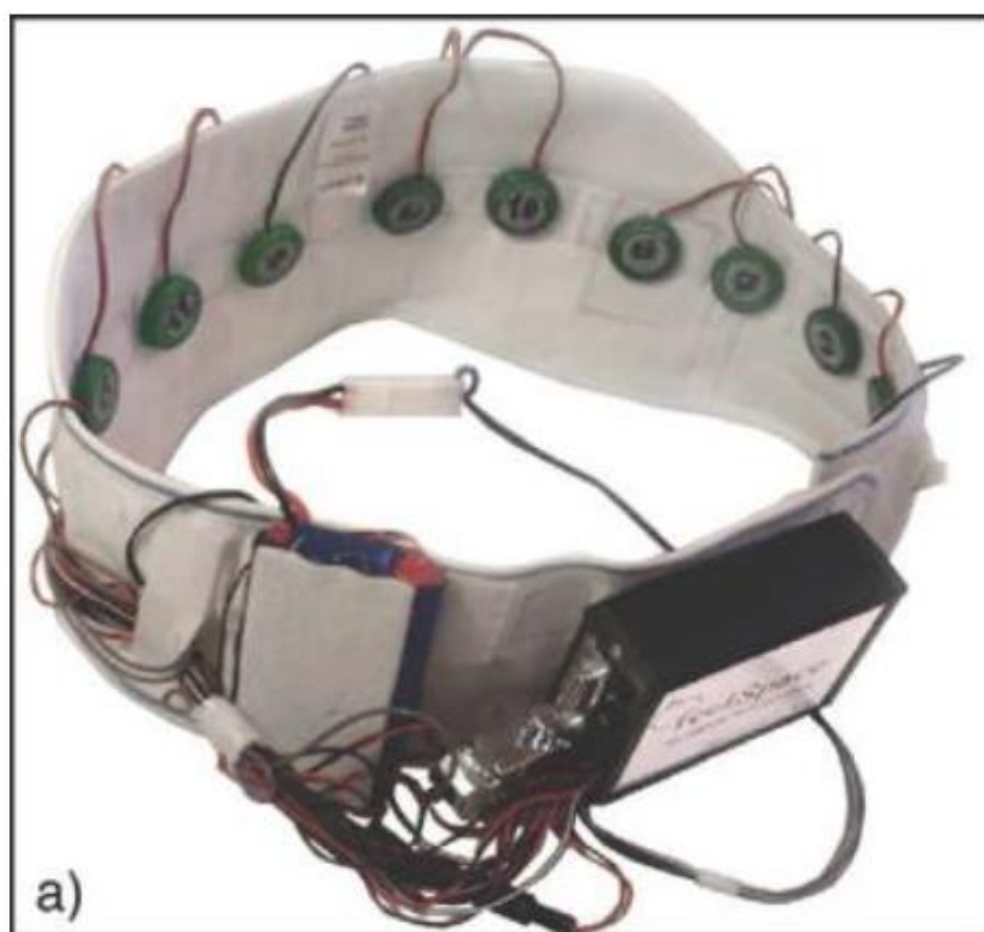
Artificial general intelligence (AGI)



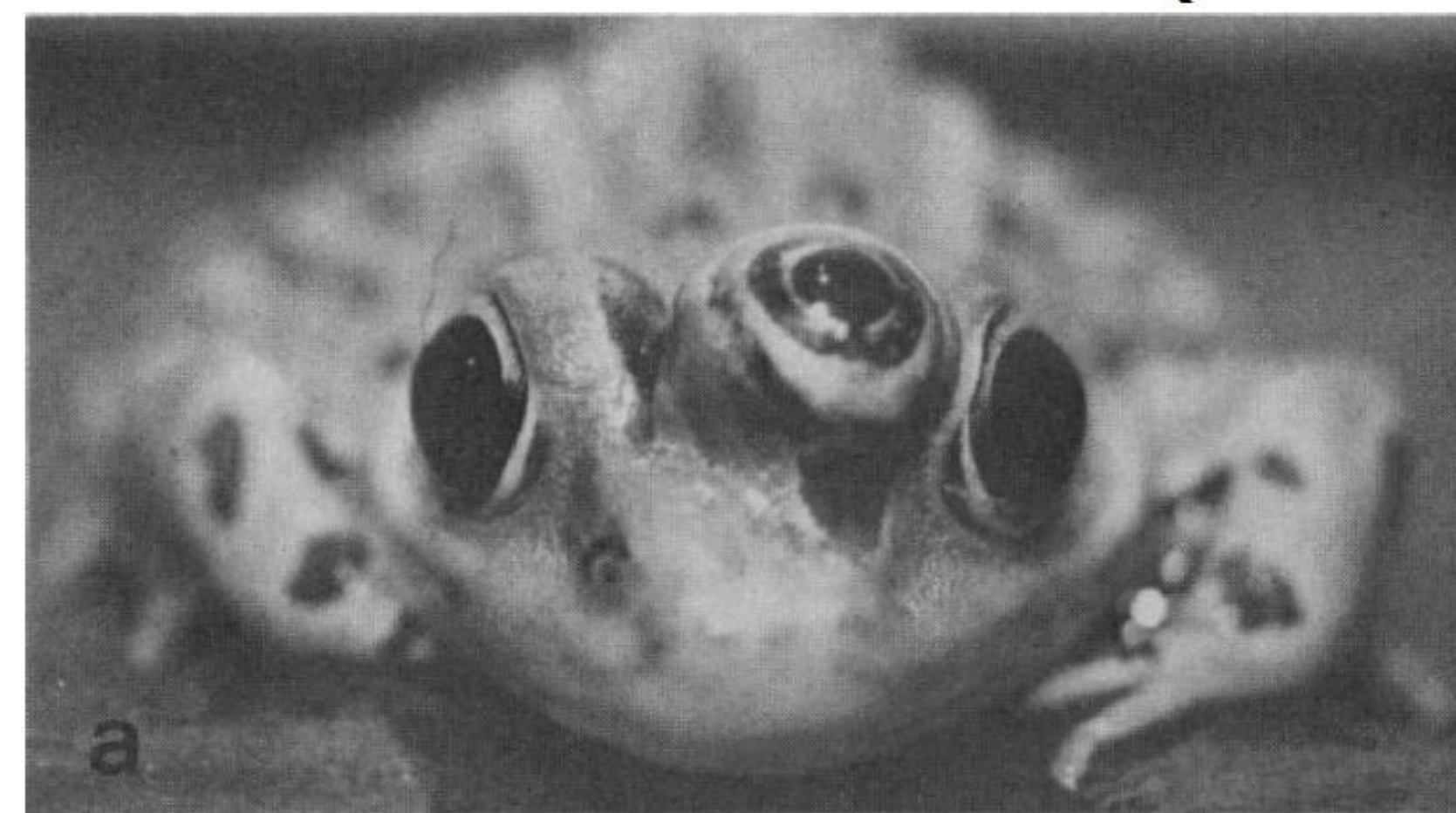
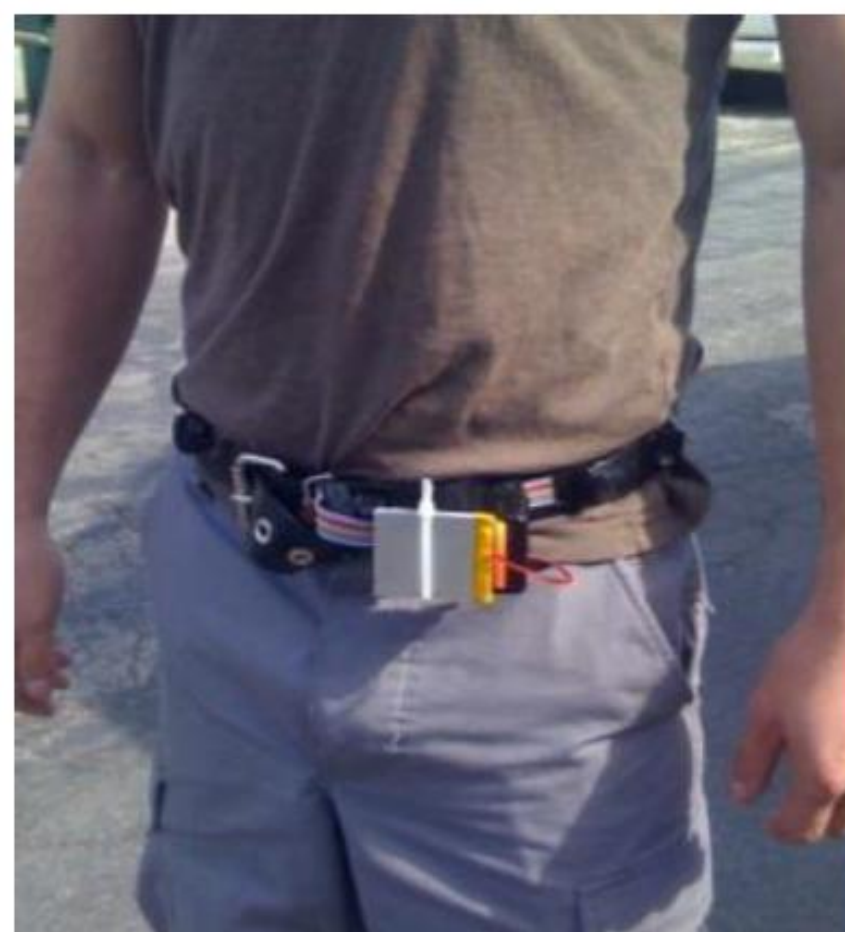
Seeing with your tongue



Human echolocation (sonar)



Haptic belt: Direction sense



Implanting a 3rd eye

Thank You

Any questions?

kushol@iut-dhaka.edu

Source & Special Thanks to Andrew Ng (Coursera)
Machine Learning Course 2

<https://www.youtube.com/@Deeplearningai>

<https://www.youtube.com/watch?v=ggWLvh484hs&list=PLyoNSC4BT4eVpykPF0Yx8C1Zs50XtD17L>